Homotopical Encryption: Hidden Hodge Operators and the Harmonic Cochain Problem

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Abstract

We introduce: (i) the Harmonic Cochain Problem (HCP), the problem of recovering $h \in \ker(\Delta_1)$ from public data (α, b), where Δ_1 is the degree-one Hodge–Laplacian induced by some hidden representative $x \in A_\alpha = \pi^{-1}(\alpha)$ of the public homotopy class label α. This gives adjacent coboundaries δ_0, δ_1, the degree-one Hodge–Laplacian Δ_1, the Hodge projection π_H, and the Hodge-structured decomposition $b = h+e$ with $e \in \mathrm{im}(\delta_0) \oplus \mathrm{im}(\delta_1^*)$; (ii) a CVP comparison that separates the field-linear image $\mathcal{V}_B(R) = \{Bx : x \in R^n\}$ from the discrete lattice $\Lambda_B = B\mathbb{Z}^n$: the former can be realized as a linear kernel, while for an explicitly supplied inclusion $\Lambda_B \subseteq W$, orthogonal projection onto the harmonic subspace W preserves the closest lattice vectors; (iii) a hidden-operator separation: a correct known-lattice CVP oracle does not uniformly solve public-data HCP when publicly equivalent hidden operators may disagree on harmonicity; (iv) the supporting formal framework, including fibrational identity data, homotopy classes, exact Hodge/cochain invariants, spectral/Tsirelson admissibility, and Heyting-style oracle judgments; and (v) a public projection chain through Plucker coordinates and cochain commitments to a Reed–Solomon mutual-correlated-agreement endpoint, conditional on an explicit far-separation witness and a cited coding-theory bound. The definitions, theorem boundaries, and Lean formalization are explicitly indexed in the appendices.

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1 Introduction

1.1 Motivation And Contributions

Homotopical Encryption studies recovery from public homotopy-class data when the Hodge operator whose kernel is the message space is not part of the public input. In a known-lattice closest-vector problem, the lattice is presented before the search begins. In HCP, the recovery target is

$$h = \pi_H(b) \in \ker(\Delta),$$

but the public class does not include the Hodge–Laplacian Δ, its kernel, or the restriction maps from which Δ is built. The private recovery object is therefore not a scalar secret or a vector in a published lattice; it is the harmonic cochain in the kernel of a hidden Hodge operator.

The operator dependence can be summarized as follows. A hidden representative determines the restriction data, which determines the adjacent coboundaries and hence the degree-one Hodge operator:

$$x \longmapsto \{R^x_{v\to e}\} \longmapsto (\delta_{0,x}, \delta_{1,x}) \longmapsto \Delta_{1,x} \longmapsto \big(\ker \Delta_{1,x}, \pi_{H_x}\big).$$

The recovery operation is Hodge-theoretic; "homotopical" describes the framework in which multiple hidden representatives may occupy the same public class while inducing different operators. The distinction is important: it is hidden-operator ambiguity, not whether a displayed matrix happens to have binary entries, that drives HCP.

This changes the comparison problem. A real linear span can be realized as the kernel of a suitable linear operator, but a discrete lattice cannot equal a nonzero real kernel. If an explicit lattice Λ is supplied inside a harmonic subspace $W = \ker(\Delta_W)$, orthogonal projection of the target onto W preserves the closest vectors of Λ. That bridge does not make public-data HCP a known-lattice problem: public homotopy-class data may be compatible

with distinct hidden operators, distinct kernels, and different harmonicity judgments. The paper isolates this boundary and then separates it from the finite verifier-facing projection artifacts used later. Those artifacts supply concrete sources for the abstract observation (α, b): the public homotopy class supplies the realized class label α, while accepted cochain payloads, openings, or proof-carrying statements supply the cochain-facing datum b. They bind to hidden Hodge data without presenting Δ or $\ker(\Delta)$. The fibration ID is the spectral fingerprint $\mathrm{Spec}_{\mathrm{multi}}(\Delta)$, obtained only after the relevant Hodge data has been recovered or certified; it is not part of the public homotopy class.

The principal notation is fixed globally as follows. Symbols used only inside a single proposition or construction are defined locally.

Symbol	Meaning
$p : E \to B$	Helfer-style fibration projection
$\mathcal{P}_X(x, y)$	path object from x to y in X
$\pi : A \to P$	public homotopy-class classifier
$\alpha = \pi(x) \in P$	realized public class label or fingerprint
$A_\alpha = [x]_\pi$	fiber of hidden representatives with public label α
(α, b)	public HCP observation
$\delta_0, \delta_1, \Delta_1$	adjacent coboundaries and degree-one Hodge–Laplacian
$\ker(\Delta_1),\ \pi_H$	harmonic subspace and harmonic projection

Table 1: Core notation used across the semantic and HCP layers.

This paper develops this boundary through the following statements.

(i) The Harmonic Cochain Problem (HCP): recovery of a cochain h with $\Delta h = 0$ for a hidden Hodge operator Δ.

(ii) A Hodge-structured noise model with cochain decomposition $b = h + e$, where $h \in \ker(\Delta_1)$ and $e \in \mathrm{im}(\delta_0) \oplus \mathrm{im}(\delta_1^*)$.

(iii) A CVP comparison separating field-linear image from discrete lattice: generated subspaces can be realized as kernels, and, when $\Lambda \subseteq W$ is explicitly supplied, projecting the target onto W preserves the discrete closest-vector condition on Λ.

(iv) A hidden-operator ambiguity theorem: publicly equivalent Hodge operators may disagree on harmonicity, so the CVP comparison does not make a correct explicit-lattice CVP oracle into a uniform public-data HCP solver.

(v) The supporting formal framework: fibrational identity, homotopy classes, exact Hodge/cochain invariants, spectral/Tsirelson admissibility, and Heyting-style public judgments.

(vi) A projection chain through Plucker coordinates and cochain commitments to a canonical Reed–Solomon endpoint, with a conditional far-separation gate and an MCA bad-challenge bound imported at an explicit cited theorem boundary.

The corresponding Lean definitions, reductions, gates, theorem boundaries, paper-to-Lean cross-reference, and file-by-file source coverage are recorded in the appendices.

2 The Harmonic Cochain Problem

Let A denote the type of hidden representatives, let P denote the type of public class data, and let

$$\pi : A \to P, \qquad \alpha \in \mathrm{im}(\pi), \qquad A_\alpha = \pi^{-1}(\alpha) = \{x \in A : \pi(x) = \alpha\}.$$

For $x \in A$, we also write $[x]_\pi := A_{\pi(x)}$. Thus, whenever $\alpha = \pi(x)$, the two descriptions agree: $A_\alpha = [x]_\pi$.

Definition 2.1 (Harmonic Cochain Problem)**.** *The* Harmonic Cochain Problem *(HCP) asks, given public data (α, b) consisting of a realized public homotopy-class label $\alpha \in \mathrm{im}(\pi)$ and a cochain-facing observation b, to recover a cochain $h \in \ker(\Delta_1)$, where Δ_1 is the degree-one Hodge–Laplacian induced by some hidden representative $x \in A_\alpha = \pi^{-1}(\alpha)$.*

In the Homotopical Encryption public-sample model, the visible cochain has the form

$$b = h + e, \qquad h \in \ker(\Delta_1), \qquad e \in \mathrm{im}(\delta_0) \oplus \mathrm{im}(\delta_1^*).$$

Here h is the degree-one harmonic cochain, equivalently a harmonic representative of its degree-one cohomology class, and e is Hodge-structured noise. The hidden-operator condition is that the public data does not determine a unique Δ_1 or $\ker(\Delta_1)$. The algebraic recovery chain is

$$x \longmapsto \{R_{v\to e}\} \longmapsto (\delta_0, \delta_1) \longmapsto \Delta_1 \longmapsto \pi_H : C^1 \to \ker(\Delta_1),$$

where $x \in A_\alpha$ is a hidden representative, $R_{v\to e}$ are the restriction maps determining the adjacent coboundaries

$$\delta_0 : C^0 \to C^1, \qquad \delta_1 : C^1 \to C^2,$$

and

$$\Delta_1 = \delta_0\delta_0^* + \delta_1^*\delta_1$$

is the induced degree-one Hodge–Laplacian. The map π_H is the orthogonal projection onto $\ker(\Delta_1)$. When the degree is fixed, we write Δ for this operative Laplacian. The public observation may bind to this chain, but does not expose the chain.

Definition 2.2 (Matrix harmonicity)**.** *For a hidden linear Hodge operator Δ, a cochain $h \in R^N$ is Δ-harmonic when*

$$\Delta h = 0.$$

Thus the harmonic kernel is $\ker(\Delta)$*. In the projection model this agrees with the condition $h \in W$ when $\Delta = \Delta_W = I - P_W$.*

Definition 2.3 (Projection Laplacian, harmonic projection, and residual)**.** *Let $V = R^n$ be equipped with its standard inner product, and let $W \subset V$ be a linear subspace. Write*

$$P_W : V \to W \subset V \qquad \textit{and} \qquad P_{W^\perp} : V \to W^\perp \subset V$$

for the orthogonal projections. The projection-model Laplacian attached to W is

$$\Delta_W = I - P_W.$$

For $b \in V$, define the harmonic projection, residual vector, and residual score by

$$\pi_W(b) = P_W b, \qquad r_W(b) = \Delta_W b, \qquad \rho_W(b) = \|r_W(b)\|.$$

We say that b is W-harmonic when $r_W(b) = 0$.

Proposition 2.4 (Projection harmonicity equivalences)**.** *For every $b \in V$,*

$$r_W(b) = P_{W^\perp} b, \qquad \rho_W(b) = \|P_{W^\perp} b\|, \qquad \rho_W(b) \geq 0.$$

Moreover

$$\pi_W(b) \in W, \qquad r_W(\pi_W(b)) = 0,$$

and the following conditions are equivalent:

$$r_W(b) = 0 \quad \iff \quad \rho_W(b) = 0 \quad \iff \quad b \in W.$$

Proof. Since $V = W \oplus W^\perp$ orthogonally, every $b \in V$ decomposes uniquely as

$$b = P_W b + P_{W^\perp} b.$$

Hence

$$(I - P_W)b = b - P_W b = P_{W^\perp} b,$$

giving $r_W(b) = P_{W^\perp} b$ and the displayed formula for $\rho_W(b)$. Nonnegativity is the nonnegativity of the norm. The projection $P_W b$ lies in W, so its orthogonal component is zero; therefore $r_W(\pi_W(b)) = 0$. Finally, $r_W(b) = 0$ iff $P_{W^\perp} b = 0$, which holds iff $b \in W$. The equivalence with $\rho_W(b) = 0$ follows from $\|x\| = 0$ iff $x = 0$. □

Proposition 2.5 (Hodge decomposition surface)**.** *Let the three consecutive cochain spaces be finite-dimensional real inner product spaces, and let the displayed stars denote adjoints. Suppose*

$$d_{\text{next}} d_{\text{prev}} = 0, \qquad \Delta = d_{\text{prev}} d_{\text{prev}}^* + d_{\text{next}}^* d_{\text{next}}.$$

Then every cochain b admits a decomposition

$$b = h + e_1 + e_2$$

with

$$\Delta h = 0, \qquad e_1 \in \operatorname{im}(d_{\text{prev}}), \qquad e_2 \in \operatorname{im}(d_{\text{next}}^*),$$

and

$$\langle h, e_1 \rangle = \langle h, e_2 \rangle = \langle e_1, e_2 \rangle = 0.$$

Proof. Since the ambient cochain space is finite-dimensional and Δ is self-adjoint, decompose it as

$$\ker(\Delta) \oplus \operatorname{im}(\Delta).$$

Since

$$\operatorname{im}(\Delta) \subseteq \operatorname{im}(d_{\text{prev}}) + \operatorname{im}(d_{\text{next}}^*),$$

every cochain b can be written as $h + e_1 + e_2$ with $h \in \ker \Delta$, $e_1 \in \operatorname{im}(d_{\text{prev}})$, and $e_2 \in \operatorname{im}(d_{\text{next}}^*)$. From $\Delta h = 0$, one obtains

$$\|d_{\text{prev}}^* h\|^2 + \|d_{\text{next}} h\|^2 = 0,$$

hence $d_{\text{prev}}^* h = 0$ and $d_{\text{next}} h = 0$. These identities give $\langle h, e_1 \rangle = 0$ and $\langle h, e_2 \rangle = 0$, while $d_{\text{next}} d_{\text{prev}} = 0$ gives $\langle e_1, e_2 \rangle = 0$. □

2.1 Public And Hidden Data

Definition 2.6 (Public observation)**.** *A public observation is the data exposed to a verifier or adversary without revealing a hidden representative. In the abstract model it consists of a realized public homotopy-class label α, together with a cochain-facing observation b. In the Homotopical Encryption instantiation, the fully public handles are hashes, roots, topology hashes, nullifiers, and bounded public summaries; cochain commitments and proof-carrying cochain payloads sit in the verifier-facing artifact layer; full cochains and openings remain witness data unless an explicit protocol path exposes them.*

Thus (α, b) is the protocol-independent abstraction of the concrete artifact-and-commitment layer in Section 5: α records the public homotopy-class label, while b records the cochain-level datum made available through an accepted payload, commitment, opening, or proof-carrying statement. This abstraction is the public datum used in the HCP definition and in the hidden-kernel ambiguity comparison.

The public observation is not the hidden Hodge system. It classifies a public equivalence class of possible hidden representatives. Fibrational and homotopical semantics enter through this distinction: public equality is equality after applying a public classifier, not strict equality of hidden representatives. Section 4 develops this semantic layer.

Definition 2.7 (Hidden-operator public projection)**.** *A public projection is a map*

$$\pi_\Delta : \{\textit{hidden Hodge operators } \Delta\} \longrightarrow P$$

to a public class type P. Two hidden operators are publicly equivalent when $\pi_\Delta(L) = \pi_\Delta(L')$. The hidden operator fiber over $\alpha \in P$ is

$$\mathcal{H}_\alpha = \pi_\Delta^{-1}(\alpha) = \{\Delta : \pi_\Delta(\Delta) = \alpha\},$$

and the hidden class of Δ is $\mathcal{H}_{\pi_\Delta(\Delta)}$.

Definition 2.8 (Lossiness and kernel ambiguity)**.** *The public projection π_Δ is* lossy *if there exist $L \neq L'$ with $\pi_\Delta(L) = \pi_\Delta(L')$. The two operators have the same harmonic kernel if*

$$\forall v, \quad Lv = 0 \Longleftrightarrow L'v = 0.$$

The projection is kernel-ambiguous *if there are publicly equivalent operators L, L' whose harmonic kernels are not the same.*

Definition 2.9 (Public-compatible Hodge ambiguity)**.** *Public-compatible Hodge ambiguity means that there exist publicly equivalent operators L, L' and a cochain-facing observation b such that*

$$Lb = 0 \qquad \textit{but} \qquad L'b \neq 0.$$

Proposition 2.10 (Ambiguity hierarchy)**.** *Public-compatible Hodge ambiguity implies kernel ambiguity, and kernel ambiguity implies that the public projection is lossy.*

Proof. If b is harmonic for L but not for L', then the two operators cannot have the same harmonic kernel. This gives kernel ambiguity. If publicly equivalent operators have different kernels, they cannot be the same operator, since equality of operators implies equality of kernels. Therefore the public projection identifies two distinct hidden operators. □

Definition 2.11 (Hidden Hodge witness)**.** *A hidden Hodge witness for a public observation (α, b) consists of hidden operator data Δ, a harmonic cochain h, and Hodge-structured noise e, such that*

$$\Delta \in \mathcal{H}_\alpha, \qquad h \in \ker(\Delta), \qquad e \in \operatorname{im}(\delta_0) \oplus \operatorname{im}(\delta_1^*), \qquad b = h + e.$$

The hidden data includes the fibration representative, restriction maps, the induced Laplacian, its harmonic kernel, and private witness/cochain data. Public data may bind to this hidden data through commitments or proof-carrying artifacts, but it does not present the operative kernel.

Definition 2.12 (HCP solution)**.** *A solution to an HCP instance (α, b) is a cochain h' that is the harmonic component of the operative hidden witness. In the exact recovery form used by the hidden-operator separation, $h' = h$ for every hidden witness (Δ, h, e) compatible with the public observation; hence uniform public recovery requires the observation to determine the relevant hidden harmonicity data.*

2.2 Recovery Target

The recovery target is the harmonic cochain in the operative kernel, not the entire kernel subspace. Thus CVP asks for a closest vector in a presented lattice, while HCP asks for a harmonic cochain in the operative Hodge kernel.

In the metric variant used for comparison with CVP, the continuous target is the closest harmonic cochain to b. Presenting Δ or a basis for $\ker(\Delta)$ presents a real subspace, not a discrete lattice. A genuine CVP instance

additionally requires an explicit lattice $\Lambda \subseteq \ker(\Delta)$; Theorem 3.2 shows that projection onto the kernel preserves the closest vectors of that supplied lattice. The hidden-operator model withholds the operative subspace itself.

2.3 Computational Model

The computational model separates three layers.

(i) The mathematical layer defines public projections, hidden operator fibers, harmonicity, and kernel ambiguity.

(ii) External hardness assumptions and cited coding-theory theorems enter at explicitly labeled theorem boundaries.

(iii) Runtime requirements such as byte parsing, canonical encodings, hash domain separation, proof verification, and state transition are outside the mathematical HCP definition.

This paper treats the first two layers. The runtime layer belongs to the subsequent protocol paper.

Lean Cross-Reference

The definitions and theorem surfaces in this section correspond to `Hardness.MatrixIsHarmonic`, `Hardness.PublicProjection`, `Hardness.HiddenOperatorFiber`, `Hardness.HiddenOperatorClass`, `Hardness.SameKernel`, `Hardness.KernelAmbiguous`, `Hardness.PublicCompatibleHodgeAmbiguity`, `Hardness.Hodge_Decomposition_Theorem`, `Hardness.kernelAmbiguous_implies_lossy`, `Hardness.kernelAmbiguous_of_publicCompatibleHodgeAmbiguity`, and `Hardness.Hodge_Ambiguity`.

3 CVP Comparison And Kernel Ambiguity

CVP asks for a closest vector in an explicitly presented lattice. HCP asks for the harmonic cochain in the kernel of a hidden Hodge operator. In the Homotopical Encryption model, public data need not determine the corresponding harmonic kernel.

3.1 Known-Lattice CVP

An explicit Euclidean CVP instance consists of a full-column-rank generator matrix $B \in \mathbb{R}^{m \times n}$ and a target $t \in \mathbb{R}^m$. Its discrete lattice is

$$\Lambda_B = B\mathbb{Z}^n = \{Bz : z \in \mathbb{Z}^n\}.$$

The injectivity of $z \mapsto Bz$ records that the displayed columns are a lattice basis. A correct closest-vector answer is a vector $v \in \Lambda_B$ such that

$$\|t - v\|^2 \leq \|t - w\|^2 \qquad \text{for every } w \in \Lambda_B.$$

Thus, for fixed dimensions, a CVP oracle O is correct when

$$\begin{aligned} \mathsf{CVP}_{\mathrm{corr}}(O) \quad \Longleftrightarrow \quad & \forall \text{ full-column-rank } B, t, \\ & O(B,t) \in \Lambda_B \quad \text{and} \quad \forall w \in \Lambda_B, \\ & \|t - O(B,t)\|^2 \leq \|t - w\|^2. \end{aligned}$$

The weaker output condition is merely lattice membership:

$$\forall B, t,\ O(B,t) \in \Lambda_B.$$

Fix a class $\mathcal{C}$ of admissible algorithms, such as polynomial-time or probabilistic polynomial-time algorithms. The external CVP hardness assumption is the schema

$$\mathsf{CVP}_{\mathrm{hard}}(\mathcal{C}) \quad \Longleftrightarrow \quad \neg\exists O,\ O \in \mathcal{C} \wedge \mathsf{CVP}_{\mathrm{corr}}(O).$$

This is the separation between the known-lattice interface and the hidden-operator interface: a CVP oracle receives B and t, while a public HCP procedure sees only public class values, hashes, commitments, and verifier-facing artifacts unless extra hidden-operator information is supplied.

3.2 Presented Linear Images

The field-linear image of B is a different object:

$$\mathcal{V}_B(R) = BR^n = \{Bx : x \in R^n\}.$$

Over a field this is a linear subspace, not the discrete lattice $B\mathbb{Z}^n$. The following comparison is therefore linear algebra; it is not a reduction from discrete CVP.

Let B present $\mathcal{V}_B(R)$. If a matrix Δ satisfies $\ker(\Delta) = \mathcal{V}_B(R)$, then any solver satisfying

$$\Delta S(\Delta, t) = 0$$

returns $S(\Delta, t) \in \mathcal{V}_B(R)$.

Proposition 3.1 (Kernel realization for explicit generated subspaces)**.** *Let $B \in R^{n\times n}$ be an explicit square generator matrix over a field and set $\mathcal{V}_B(R) = \{Bx : x \in R^n\}$. There exists a matrix Δ such that*

$$\ker(\Delta) = \mathcal{V}_B(R).$$

Consequently, any solver $S(\Delta, b)$ whose output is Δ-harmonic for every presented Δ returns a vector in the explicit generated subspace $\mathcal{V}_B(R)$ when run on this realized operator.

Proof. Let $B \in R^{n\times n}$ be an explicit generator matrix over a field R, set

$$\mathcal{V}_B(R) = \{Bx : x \in R^n\} \subseteq R^n,$$

and let $t \in R^n$ be the solver input. We construct a matrix Δ with

$$\ker(\Delta) = \mathcal{V}_B(R).$$

Put $V = \mathcal{V}_B(R)$. Since R^n is finite-dimensional over a field, choose a complementary subspace W with

$$V + W = R^n, \qquad V \cap W = 0.$$

Thus every $y \in R^n$ has a unique decomposition

$$y = v + w, \qquad v \in V, \quad w \in W.$$

Equivalently, there is a unique element $w_y \in W$ such that $y - w_y \in V$. Define

$$P(y) = w_y.$$

The uniqueness of the $V \oplus W$ decomposition gives linearity: for $y, z \in R^n$, the vector $P(y) + P(z)$ lies in W, and

$$(y + z) - (P(y) + P(z)) = (y - P(y)) + (z - P(z)) \in V,$$

so $P(y + z) = P(y) + P(z)$. Similarly, for $a \in R$, $aP(y) \in W$ and

$$ay - aP(y) = a(y - P(y)) \in V,$$

so $P(ay) = aP(y)$.

We now compute the kernel. If $y \in V$, then $0 \in W$ and $y - 0 \in V$, hence uniqueness gives $P(y) = 0$. Conversely, if $P(y) = 0$, then $y - P(y) = y \in V$. Therefore

$$\ker P = V = \mathcal{V}_B(R).$$

Let Δ be the matrix of the linear map P. Then, for every $y \in R^n$,

$$\Delta y = 0 \quad \Longleftrightarrow \quad y \in \ker P \quad \Longleftrightarrow \quad y \in \mathcal{V}_B(R) \quad \Longleftrightarrow \quad \exists x,\ Bx = y.$$

Let $S : R^{n\times n} \times R^n \to R^n$ be a presented-operator Hodge solver satisfying

$$\forall \Delta\, b, \qquad \Delta(S(\Delta, b)) = 0.$$

Applying this to the Δ constructed above and $b = t$ gives

$$\Delta(S(\Delta, t)) = 0.$$

By the kernel calculation, there exists x with

$$Bx = S(\Delta, t).$$

Hence $S(\Delta, t) \in \mathcal{V}_B(R)$. □

When $R = \mathbb{R}$, if a presented-operator solver additionally minimizes Euclidean distance among Δ-harmonic vectors, the same kernel equality says that it returns the continuous closest point of $\mathcal{V}_B(\mathbb{R})$. This is still not discrete CVP unless a separate lattice is supplied.

3.3 Explicit Harmonic-Lattice Bridge

The discrete bridge requires both kinds of data: a real harmonic subspace and an integer lattice contained in it. Let $W \subseteq \mathbb{R}^m$ be the operative harmonic subspace, let P_W be its orthogonal projection, and let $\Lambda_B = B\mathbb{Z}^n \subseteq W$ be an explicitly presented lattice. Set

$$\Delta_W = I - P_W.$$

The inclusion $\Lambda_B \subseteq W$ does not identify the discrete lattice with the real subspace.

Proposition 3.2 (Discrete CVP through harmonic projection)**.** *For $t \in \mathbb{R}^m$, put $t_W = P_W t$. Then:*

(i) t_W and every $\lambda \in \Lambda_B$ are Δ_W-harmonic;

(ii) for every $\lambda \in \Lambda_B$,

$$\|t - \lambda\|^2 = \|t - t_W\|^2 + \|t_W - \lambda\|^2;$$

(iii) a lattice vector λ is closest to t in Λ_B if and only if it is closest to t_W in Λ_B.

Proof. Since $t_W \in W$, every $\lambda \in \Lambda_B$ also lies in W, and $\ker(\Delta_W) = W$, the harmonicity claims follow. For the distance identity, write

$$t - \lambda = (t - t_W) + (t_W - \lambda).$$

Orthogonal projection gives $t - t_W \in W^\perp$, while $t_W - \lambda \in W$. The two summands are orthogonal, so the Pythagorean identity gives (ii). The term $\|t - t_W\|^2$ is independent of λ; consequently, for any $\lambda, \mu \in \Lambda_B$,

$$\|t - \lambda\|^2 \leq \|t - \mu\|^2 \quad \Longleftrightarrow \quad \|t_W - \lambda\|^2 \leq \|t_W - \mu\|^2.$$

This is exactly (iii). □

The continuous projection $t_W = P_W t$ is the closest harmonic vector to t, but it need not lie in Λ_B. Thus projection transports the target to the harmonic subspace while preserving the discrete minimizers; it does not perform the remaining lattice search. The proposition also presumes that W and the inclusion $\Lambda_B \subseteq W$ have been supplied. It does not discover the operative hidden subspace from public class data.

3.4 Oracle Insufficiency

The oracle-insufficiency direction uses a different contract. A public Hodge solver may call an explicit-lattice CVP oracle, but it receives no hidden operator. The conditions below require uniform harmonicity with respect to every hidden operator and fixedness on inputs already harmonic for that operator.

Definition 3.3 (Public-data Hodge solver contract)**.** *Let*

$$S : \mathcal{O}_{\mathrm{CVP}} \times \mathbb{R}^n \longrightarrow \mathbb{R}^n,$$

where $\mathcal{O}_{\mathrm{CVP}}$ is the class of explicit-lattice CVP oracles. For $O \in \mathcal{O}_{\mathrm{CVP}}$, define

$$\mathsf{H}_O(S) \Longleftrightarrow \forall \text{ linear } L : \mathbb{R}^n \to \mathbb{R}^n, \quad \forall b \in \mathbb{R}^n, \quad L(S(O, b)) = 0,$$
$$\mathsf{F}_O(S) \Longleftrightarrow \forall \text{ linear } L : \mathbb{R}^n \to \mathbb{R}^n, \quad \forall b \in \mathbb{R}^n, \quad Lb = 0 \Rightarrow S(O, b) = b.$$

The public-data-only contract is

$$\exists S, \quad \forall O, \ \mathsf{CVP}_{\mathrm{corr}}(O) \Rightarrow (\mathsf{H}_O(S) \wedge \mathsf{F}_O(S)).$$

Proposition 3.4 (Public ambiguity and CVP oracle insufficiency)**.** *Suppose $n \geq 2$. Then:*

(i) The constant public projection on hidden linear Hodge operators is public-compatible Hodge ambiguous, hence kernel-ambiguous and lossy.

(ii) If at least one correct explicit-lattice CVP oracle exists, namely

$$\exists O,\ \mathsf{CVP}_{\mathrm{corr}}(O),$$

then

$$\neg\exists S, \quad \forall O,\ \mathsf{CVP}_{\mathrm{corr}}(O) \Rightarrow (\mathsf{H}_O(S) \wedge \mathsf{F}_O(S)).$$

Hence a correct oracle for explicit known-lattice CVP does not by itself determine the operative Hodge–Laplacian kernel.

Proof. Assume $n \geq 2$, and choose an index i. Let

$$b = e_i, \qquad L = 0, \qquad L' = \mathrm{diag}(1, 0, \ldots, 0),$$

where the nonzero diagonal entry of L' is placed at i. Then

$$Lb = 0, \qquad L'b = b \neq 0.$$

Both operators are self-adjoint positive-semidefinite projection-model Laplacians: $L = I - P_{\mathbb{R}^n}$, while $L' = I - P_{e_i^\perp}$. Thus b is harmonic for L but not harmonic for L'. The constant public projection sends every hidden operator to the same public value, so L and L' are publicly equivalent. The same cochain-facing observation b is therefore compatible with two hidden Hodge operators that disagree about whether b lies in the harmonic kernel. This gives public-compatible Hodge ambiguity; the implications of Theorem 2.10 give kernel ambiguity and lossiness.

Now suppose, in addition, that $\exists O, \mathsf{CVP}_{\mathrm{corr}}(O)$. Suppose, for contradiction, that S satisfies the displayed public-data-only contract. Choose O with $\mathsf{CVP}_{\mathrm{corr}}(O)$. Then

$$\mathsf{H}_O(S) \wedge \mathsf{F}_O(S).$$

Since $Lb = 0$, the condition $\mathsf{F}_O(S)$, applied to (L, b), gives

$$S(O, b) = b.$$

The condition $\mathsf{H}_O(S)$, applied to (L', b), gives

$$L'S(O, b) = 0.$$

Combining the two equalities gives $L'b = 0$, contradicting the construction of b. Hence no such public-data-only solver exists, even assuming the availability of a correct oracle for explicitly presented CVP instances. □

The obstruction is kernel ambiguity: the same public observation may be compatible with multiple hidden operators whose harmonic kernels differ. A CVP oracle can solve a closest-vector problem after the lattice is presented. The bridge in Theorem 3.2 applies only after the operative harmonic subspace and the lattice inclusion have also been supplied; it does not determine which hidden kernel is compatible with the public data.

Lean Cross-Reference

The discrete CVP definitions correspond to `CVP.ExplicitLattice`, `CVP.Instance`, `CVP.InExplicitLattice`, `CVP.squaredEuclideanDistance`, `CVP.IsClosestLatticeVector`, `CVP.Solution`, `CVP.OracleCorrect`, and `CVP.HardnessAssumption`. The separate field-linear comparison is `Hardness.exists_matrix_ker_eq_range` and `Hardness.Hodge_GeneratedSubspace_Reduction_Proof`. The explicit harmonic-lattice bridge is `Hardness.HarmonicCVPBridge`, `Hardness.HarmonicCVPBridge.squaredDistance_projection_decomposition`, `Hardness.HarmonicCVPBridge.isClosestLatticeVector_iff_projectedTarget`, and `Hardness.HarmonicCVPBridge.solution_iff_projectedTarget`. The closest-harmonic and hidden-operator surfaces are `Hardness.ClosestHarmonicSolution`, `Hardness.ProjectionClosestHarmonicSolution`, `Hardness.matrix_harmonic_iff_projection_isHarmonic`, `Hardness.no_public_solver_on_ambiguous_input`, and `Hardness.CVP_Oracle_Insufficient_For_Public_Hodge_Solving`.

4 Axioms And Semantics

4.1 Fibrational Semantics

The semantic layer is motivated by Helfer's first-order homotopical logic [Hel19]. We use the associated path, fibration, sheaf, and internal-logic language for hidden representatives, public bases, and transport of witness data; the external foundational results are taken as background rather than reproved. We use the usual fibrational notation

$$p : E \longrightarrow B, \qquad E_b = p^{-1}(b)$$

for a total space of hidden data over a base of public or contextual data. When $x, y \in E_b$, equality in the semantic layer is interpreted through a path

object rather than by strict equality in the ambient presentation. We write $\mathcal{P}_X(x, y)$ for the path object from x to y in a space or type X, and set

$$x \simeq_b y \quad \Longleftrightarrow \quad \mathcal{P}_{E_b}(x, y) \neq \varnothing.$$

For a dependent family $F : E \to \mathsf{Type}$, a path $\gamma \in \mathcal{P}_E(x, y)$ supplies the transport map

$$\gamma_* : F_x \longrightarrow F_y.$$

Thus the primitive semantic data are fibers, paths, and transport along paths. The cryptographic objects introduced later live over this interface; they do not replace it with byte-level equality.

We also use the corresponding categorical fibration vocabulary. For a functor $p : E \to B$, a morphism $f : U \to V$ in E is p-Cartesian when every arrow $\psi : W \to V$ and base arrow $u : p(W) \to p(U)$ satisfying

$$p(\psi) = p(f) \circ u$$

admit a unique factor $\widetilde{\psi} : W \to U$ with

$$p(\widetilde{\psi}) = u, \qquad f \circ \widetilde{\psi} = \psi.$$

A Grothendieck fibration is then a functor with Cartesian lifts for every base arrow. This is the categorical form of the transport principle used by the hidden-representative semantics.

Proposition 4.1 (Fibration copies preserve hidden semantics)**.** *Let $p : E \to B$ be a Grothendieck fibration. For each tag n, let $E^{(n)}$ be a tagged copy of E and let $p^{(n)} : E^{(n)} \to B$ be the induced fibration. Then $p^{(n)}$ is again a Grothendieck fibration, and p and $p^{(n)}$ are homotopy equivalent over B.*

Proof. The tagged copy has the same objects and morphisms as E, with only the external tag changed. The forgetful functor $E^{(n)} \to E$ and the tagging functor $E \to E^{(n)}$ compose to the identity up to the evident natural isomorphisms. Cartesian lifts in $E^{(n)}$ are transported directly from Cartesian lifts in E, so the induced functor remains a Grothendieck fibration. The two functors give the homotopy equivalence over the base B. □

4.2 Fibrational Identity Objects

Let $\Delta_A : A \to A \times A$ denote the diagonal. In path-object notation, evaluation at the endpoints gives

$$A^I \xrightarrow{(e_0, e_1)} A \times A, \qquad \mathcal{P}_A(x, y) = (e_0, e_1)^{-1}(x, y).$$

Following the path/diagonal semantics, the identity object over a pair (x, y) has fiber $\mathcal{P}_A(x, y)$. We therefore regard x and y as equal for the first-order homotopical semantics exactly when this fiber is inhabited:

$$x \simeq_A y \quad \Longleftrightarrow \quad \mathcal{P}_A(x, y) \neq \varnothing.$$

When path objects carry identity paths, reversal, and concatenation, the relation $\simeq_A$ is reflexive, symmetric, and transitive. For structures M, N in the same homotopical semantic class, the corresponding invariance principle is

$$M \simeq N \quad \Longrightarrow \quad (M \models \varphi \Longleftrightarrow N \models \varphi) \quad \text{for every first-order formula } \varphi.$$

This is the semantic source of the hidden-operator model used in HCP: the verifier may receive evidence that two representatives lie in the same homotopical identity class without receiving the representative or the path witness itself.

4.3 Homotopy Classes

Let A be a type of hidden representatives and let P be a type of public class data. A public classifier is a map

$$\pi : A \longrightarrow P.$$

It determines public fibers

$$A_\alpha = \pi^{-1}(\alpha) = \{x \in A : \pi(x) = \alpha\}, \qquad \alpha \in \mathrm{im}(\pi),$$

and a public equivalence relation

$$x \equiv_\pi y \quad \Longleftrightarrow \quad \pi(x) = \pi(y).$$

The classifier is path-respecting when

$$x \simeq_A y \Longrightarrow x \equiv_\pi y,$$

and non-faithful when some public fiber contains distinct hidden representatives. A hidden class-membership witness is a path $\gamma \in \mathcal{P}_A(x, y)$; its public consequence is only the equality $\pi(x) = \pi(y)$. At this stage the public datum is only the homotopy-class value of the classifier. The finite-code projection and Reed–Solomon/MCA endpoint are introduced later in Section 6.

The fibration projection and public classifier describe compatible but distinct levels. When the hidden representatives are the objects of E, a map $\kappa : B \to P$ on objects supplies the compatibility condition

$$\pi = \kappa \circ p.$$

Then $p(x)$ is the fibration-base point of x, while $\alpha = \pi(x) = \kappa(p(x))$ is its public class label. The special case $B = P$ and $\kappa = \mathrm{id}_B$ identifies the underlying object maps π and p. In general, κ may be lossy, and

$$A_\alpha = \bigcup_{\substack{b \in B \\ \kappa(b) = \alpha}} E_b.$$

This compatibility datum is additional structure; it is not implied merely by the requirement that π respect paths.

4.4 Axiom Interfaces

The hypotheses used below are grouped by mathematical interface. This grouping does not assert that every item below is literally a file under `Axioms/`. The relevant interfaces are the following.

(i) Foundational cochain/Hodge interface: a cochain complex, a coboundary, the induced Hodge–Laplacian, Hodge decomposition, and orthogonality.

(ii) Hidden semantic interface: paths, transport, fibration structure, homotopy classes, public classifiers, and the nonfaithfulness of public class data.

(iii) Sheaf-coherence interface: restriction-shape compatibility, naturality, $d^{k+1}d^k = 0$, the induced Hodge–Laplacian identity, finiteness, simpliciality, and connectedness.

(iv) Spectral and stage-admissibility interface: Tsirelson-style spectral window conditions, near-boundary conditions, and object-local admissibility wrappers for candidate Hodge–Laplacian data.

(v) Public-judgment interface: finite Heyting collapse, public truth equivalence, oracle query/answer conditions, and oracle-visible spectral, identity, and topology judgments.

(vi) Public-artifact interface: cochain-commitment payloads, commitment security assumptions, proof-system assumptions, proof-carrying artifacts, and compression/observation boundaries used in Section 5.

The well-formedness boundaries are displayed below as enumerated conditions.

Definition 4.2 (Helfer semantic boundary)**.** *A first-order homotopical semantic datum is admissible if and only if all of the following hold:*

(i) Equality is interpreted by the path/diagonal condition $x \simeq_A y \iff \mathcal{P}_A(x, y) \neq \varnothing$.

(ii) Cartesian lifting is available for the relevant fibrations.

(iii) Dependent sums and products are coherent with transport and base-change along the fibration.

(iv) Homotopy-equivalent structures have the same first-order theory.

(v) Class membership is witnessed by hidden path data, and under a path-respecting public classifier its public consequence is $\pi(x) = \pi(y)$.

These clauses correspond to `Axioms.HelferFOL` *and* `Axioms.HelferFOLWellFormed`.

Definition 4.3 (Foundation assumption boundary)**.** *A foundation datum is admissible if and only if all of the following hold:*

(i) $C^\bullet$ *is a cochain complex.*

(ii) Each coboundary $d^k : C^k \to C^{k+1}$ *is linear.*

(iii) The Hodge–Laplacian is obtained from adjacent coboundaries and adjoints.

(iv) C^1 *admits the Hodge decomposition*

$$C^1 = \mathcal{H}^1 \oplus \operatorname{im}(\delta_0) \oplus \operatorname{im}(\delta_1^*).$$

(v) The summands $\mathcal{H}^1$, $\operatorname{im}(\delta_0)$, *and* $\operatorname{im}(\delta_1^*)$ *are pairwise orthogonal.*

(vi) A commitment collision gives an element of the relevant coboundary kernel.

(vii) The commitment hides the witness component specified by the commitment interface.

(viii) The commitment operation is additive with respect to the cochain group law.

These clauses correspond to `Axioms.Foundation` *and* `Axioms.FoundationWellFormed`*.*

Definition 4.4 (Sheaf-coherence assumption boundary)**.** *A sheaf-coherence datum is admissible if and only if all of the following hold:*

(i) Restriction maps have compatible source and target dimensions on incidences.

(ii) For every edge, the two restriction/transport paths commute.

(iii) On every triangle, the face-composition maps agree.

(iv) Adjacent sheaf coboundaries square to zero: $d^{k+1}d^k = 0$*.*

(v) The Hodge–Laplacian is obtained from the sheaf coboundaries and adjoints.

(vi) The harmonic, exact, and coexact summands are pairwise orthogonal.

(vii) The nerve is finite.

(viii) The nerve is simplicial.

(ix) The nerve is connected.

These clauses correspond to `Axioms.SheafCoherence` *and* `Axioms.SheafCoherenceWellFormed`*.*

The preceding boundary is a condition on the ambient sheaf complex. A particular serialized commitment requires an additional, element-dependent condition. Let

$$\mathrm{dec}_S : \mathcal{C}_{\mathrm{com}} \longrightarrow C_S^1$$

be the interpretation of commitment payloads in the degree-one cochain carrier of S, and let c binds S denote the topology-binding condition for that interpretation.

Definition 4.5 (Committed one-cochain coherence)**.** *A commitment payload* c *is coherent with a sheaf datum* S*, written* $c \in \mathcal{G}_S$*, if and only if*

(i) *c is well formed as a serialized commitment payload;*

(ii) *S satisfies the sheaf-coherence boundary of Theorem 4.4;*

(iii) *c* binds *S; and*

(iv) *there is an $a \in C_S^0$ such that*

$$\mathrm{dec}_S(c) = \delta_0 a.$$

Thus c represents an exact one-cochain, rather than merely having a valid wire shape.

Proposition 4.6 (Committed exact cochains are cocycles)**.** *If $c \in \mathcal{G}_S$, then*

$$\delta_1 \,\mathrm{dec}_S(c) = 0.$$

Proof. Choose $a \in C_S^0$ with $\mathrm{dec}_S(c) = \delta_0 a$. Ambient sheaf coherence gives the actual cochain-map equation $\delta_1 \delta_0 = 0$. Hence

$$\delta_1 \,\mathrm{dec}_S(c) = \delta_1 \delta_0 a = 0.$$

□

Definition 4.7 (Proof-system and commitment-security boundaries)**.** *The commitment-security boundary holds if and only if all of the following hold:*

(i) *Binding: a collision requires a nontrivial relation in the relevant coboundary kernel.*

(ii) *Hiding: the commitment distribution hides the committed witness component.*

(iii) *Homomorphic addition: $C(m_1, r_1) + C(m_2, r_2) = C(m_1 + m_2, r_1 + r_2)$ when the two commitments are compatible.*

(iv) *Homomorphic scalar multiplication: $a\,C(m, r) = C(am, ar)$ for admissible scalars a.*

(v) *Topology binding: a commitment is bound to its underlying simplicial complex or topology identifier.*

(vi) *Partial opening soundness: opening a specified subset verifies the corresponding local cochain constraints.*

The proof-system boundary holds if and only if all of the following hold:

(i) Completeness: an honest proof for a valid witness verifies.

(ii) Soundness: verifier acceptance implies validity of the asserted mathematical statement, up to the stated error model.

(iii) Zero knowledge: the proof transcript does not identify the hidden witness.

(iv) Statement binding: the proof is bound to its public statement, topology, and instance context.

These clauses correspond to `Axioms . CochainCommitment . CochainCommitmentAxioms`, `Axioms . CochainCommitment . CochainCommitmentWellFormed`, `Axioms . ZKP`, *and* `Axioms . ZKPWellFormed`.

Consensus, ledger, reliability, and Proof-of-Value records are mentioned here only at their mathematical interface boundary. Their runtime policy content belongs to the subsequent protocol paper.

4.5 Exact Structural Invariants

The exact structural invariants are not a separate source of geometry. They package the same foundational and sheaf-coherence hypotheses at the point where an object is admitted. We use the cellular-sheaf Hodge–Laplacian viewpoint of Hansen–Ghrist as background [HG18]. For a finite cell poset X, a cellular sheaf is represented here as

$$X^{\mathrm{op}} \xrightarrow{F} \mathsf{Vect}, \qquad C^0(X;F) \xrightarrow{\delta_0} C^1(X;F) \xrightarrow{\delta_1} C^2(X;F).$$

For a cochain complex $C^\bullet$, the accepted data supply differentials

$$d^k : C^k \longrightarrow C^{k+1}$$

with $d^{k+1}d^k = 0$. Together with adjoints, these give the degree-k Hodge–Laplacian

$$\Delta_k = d^{k-1}(d^{k-1})^* + (d^k)^* d^k.$$

In the sheaf setting, the corresponding coherence assumptions assert that restriction maps, local data, and gluing data are compatible with this cochain/Hodge structure. The only structural consequences used later are the adjacent differential identity $d^{k+1}d^k = 0$ and the Hodge–Laplacian identity

above. For the projection chain, we additionally require the maps used to pass from hidden sheaf data to public cochain-facing data to commute with the coboundary:

$$\begin{array}{ccc} F(V) & \xrightarrow{\eta_V} & G(V) \\ \delta_F \downarrow & & \downarrow \delta_G \\ F(E) & \xrightarrow{\eta_E} & G(E) \end{array} \qquad \eta_E \circ \delta_F = \delta_G \circ \eta_V.$$

This is the functorial/naturality diagram used in the paper. It is a cochain-level coherence condition, not a runtime statement about byte transport or consensus execution.

Proposition 4.8 (Exact structural preconditions)**.** *If a sheaf exact-gate bundle is accepted as well formed, then its sheaf-coherence data supply*

$$d^{k+1}d^k = 0 \qquad \textit{and} \qquad \Delta_k = d^{k-1}(d^{k-1})^* + (d^k)^* d^k.$$

for the cochain complex under discussion.

Proof. Well-formedness of the exact-gate bundle includes the sheaf-coherence condition. By definition of that condition, the adjacent coboundaries compose to zero and the recorded Laplacian is the displayed sum of the two coboundary terms. Projecting these two fields from the well-formedness witness gives the stated identities. □

4.6 Spectral-Gap Admissibility

The spectral condition is an object-level admissibility condition. It is not the source of HCP hardness, not a reconstruction theorem for the hidden fibration, and not introduced here as a consensus rule. In Paper 1 the PoV/Tsirelson spectral window is a mathematical condition on candidate Hodge–Laplacian data; its use as a protocol-level admissibility rule belongs to the subsequent protocol paper. The terminology is Tsirelson-style because the admissible window uses the familiar $\sqrt{2}$ scale from Tsirelson's bound [Tsi80]. For a Hermitian positive semidefinite operator Δ, write

$$\lambda_1^+(\Delta) = \inf\{\lambda : \lambda \text{ is an eigenvalue of } \Delta,\ \lambda \neq 0\},$$

with $\lambda_1^+(\Delta) = 0$ when Δ has no nonzero spectral value. The Tsirelson-style admissibility axiom is the spectral window

$$\frac{\sqrt{2}}{2} \leq \lambda_1^+(\Delta) \leq 2\sqrt{2}.$$

A separate boundary condition marks Δ as ε-near one of the two endpoints:

$$\left|\lambda_1^+(\Delta) - \frac{\sqrt{2}}{2}\right| \le \varepsilon \quad \text{or} \quad \left|\lambda_1^+(\Delta) - 2\sqrt{2}\right| \le \varepsilon.$$

This condition records near-boundary status only; runtime policy for such values belongs to the subsequent protocol paper.

The corresponding Grassmannian/Laplacian interface makes the hidden-kernel search space concrete. Throughout this subsection, set $R = \mathbb{R}$, with its standard inner product. Let

$$\mathcal{L}_\beta(R^n) = \{\Delta : \Delta = \Delta^*,\ \Delta \succeq 0,\ \dim_R \ker \Delta = \beta\}$$

be the Laplacian space at kernel dimension β. For $W \in \mathrm{Gr}(k, R^n)$, let P_W be the orthogonal projection onto W and set

$$\Delta_W = I - P_W.$$

Thus $\ker(\Delta_W) = W$, and the Grassmannian family is

$$\Omega_k = \{\Delta_W : W \in \mathrm{Gr}(k, R^n)\}.$$

Equivalently, a point of $\mathrm{Gr}(k, R^n)$ is a k-dimensional harmonic subspace W, and the map $W \mapsto \Delta_W$ sends it to the corresponding projection Laplacian. In particular, Ω_1 is the Grassmannian family of one-dimensional harmonic kernels.

Proposition 4.9 (Grassmannian Laplacian search space)**.** *For every $k \le n$, the family Ω_k is nonempty and satisfies*

$$\Omega_k \subseteq \{\Delta_W : W \subset R^n \text{ a linear subspace}\}.$$

Moreover, every element of Ω_k has a k-dimensional harmonic kernel, so $\Omega_k \subseteq \mathcal{L}_k(R^n)$.

Proof. If $k \le n$, choose k coordinate basis vectors in R^n. Their span is a k-dimensional subspace W, and $\Delta_W \in \Omega_k$, proving nonemptiness. The displayed containment is immediate from the definition of Ω_k. Since $\Delta_W = I - P_W = P_{W^\perp}$, its kernel is exactly W, hence has dimension k. □

Proposition 4.10 (Injectivity of the Grassmannian Laplacian map)**.** *If $W_1, W_2 \subset R^n$ are subspaces and*

$$\Delta_{W_1} = \Delta_{W_2},$$

then $W_1 = W_2$. Consequently $W \mapsto \Delta_W$ is injective on $\mathrm{Gr}(k, R^n)$.

Proof. The equality $\Delta_{W_1} = \Delta_{W_2}$ is the equality $I - P_{W_1} = I - P_{W_2}$, hence $P_{W_1} = P_{W_2}$. For any $x \in W_1$, $P_{W_1}x = x$, so $P_{W_2}x = x$, and therefore $x \in W_2$. Thus $W_1 \subseteq W_2$. The reverse containment is identical, so $W_1 = W_2$. □

The higher-rank Grassmannian search spaces contain explicit one-parameter families. Fix $1 \leq k < n$. Choose linearly independent vectors $u, v \in R^n$ and a $(k-1)$-dimensional subspace W_0 disjoint from $\operatorname{span}\{u, v\}$. For $t \in R$, set

$$W(t) = \operatorname{span}\{u + tv\} + W_0.$$

Then $W(t)$ has dimension k, and varying t gives a corresponding family

$$\Delta_{W_t} = I - P_{W_t}.$$

Proposition 4.11 (Parametric Laplacian search space)**.** *If $1 \leq k < n$, then there exist u, v, W_0 as above such that the map*

$$R \longrightarrow \operatorname{Gr}(k, R^n), \qquad t \longmapsto W(t)$$

is injective. Hence $t \mapsto \Delta_{W(t)}$ is an injective one-parameter family in Ω_k.

Proof. The existence of u, v, W_0 follows by choosing a $(k-1)$-plane and then two independent directions outside it; the inequality $k < n$ leaves enough room for this choice. The dimension computation uses the disjointness:

$$\dim W(t) = \dim \operatorname{span}\{u + tv\} + \dim W_0 = 1 + (k-1) = k.$$

If $W(t_1) = W(t_2)$, then $u + t_1 v$ lies in $\operatorname{span}\{u + t_2 v\} + W_0$. Modulo W_0, this gives a linear dependence between u and v unless $t_1 = t_2$. Since u, v were chosen independent and W_0 was disjoint from their span, $t_1 = t_2$. Injectivity of $t \mapsto \Delta_{W(t)}$ follows from Theorem 4.10. □

Proposition 4.12 (Projection spectrum and spectral gap)**.** *For every $W \subset R^n$, the projection Laplacian $\Delta_W = I - P_W = P_{W^\perp}$ has spectrum contained in $\{0, 1\}$. Its kernel is W, its image is $W^\perp$, and, if $W \neq R^n$, its least nonzero eigenvalue and largest eigenvalue are both 1. Thus the relative spectral gap of the Grassmannian projection family is 1.*

Proof. If $x \in W$, then $\Delta_W x = 0$. If $x \in W^\perp$, then $\Delta_W x = x$. Since $R^n = W \oplus W^\perp$, these two subspaces exhaust the eigenspace decomposition. Therefore the only possible eigenvalues are 0 and 1, with 0-eigenspace W and 1-eigenspace $W^\perp$. When $W^\perp \neq 0$, the nonzero spectral values consist only of 1, so both λ_1^+ and $\lambda_{\max}$ are 1. □

The likelihood surface records the corresponding adversary vocabulary. A classical adversary is a map from an observation b to a guessed Laplacian; a quantum adversary is a probability density on candidate Laplacians. For a noise scale σ, the Gaussian likelihood of a candidate A at observation b is written abstractly as

$$L_\sigma(b, A) = \exp\left(-\frac{\|Ab\|_2^2}{2\sigma^2}\right),$$

with normalized classical and quantum success probabilities obtained by integrating or maximizing this likelihood over the chosen family Ω_k. These likelihood objects are comparison surfaces for quantitative security statements; they are not assumptions used to prove the hidden-operator separation in Theorem 3.4.

Definition 4.13 (Adversary likelihood surfaces)**.** *A classical adversary on R^n is a function*

$$\mathcal{A}_{\mathrm{cl}} : R^n \to M_n(R)$$

from an observation to a guessed Laplacian. A quantum adversary on a candidate family Ω_k is a nonnegative density

$$\mathcal{A}_{\mathrm{q}} : R^n \times \Omega_k \to R_{\geq 0}$$

whose integral over Ω_k is 1 for each observation. With the standard Gaussian prior on observations, the corresponding normalized success quantities are obtained from

$$\int_{R^n} L_\sigma(b, \mathcal{A}_{\mathrm{cl}}(b)) e^{-\|b\|_2^2/2}\, db$$

in the classical case and

$$\int_{R^n} \left(\int_{\Omega_k} \mathcal{A}_{\mathrm{q}}(b, A) L_\sigma(b, A)\, dA \right) e^{-\|b\|_2^2/2}\, db$$

in the quantum-density case, up to the usual Gaussian normalization factor.

The likelihood is written with the Euclidean norm. The norm-convention check is that the default function-space norm may be the supremum norm, so the Euclidean likelihood must be represented by passing to the ℓ^2 norm explicitly. The mathematical surface used here is the corrected Euclidean one.

Proposition 4.14 (Likelihood for a projection Laplacian)**.** *For $A = \Delta_W$, the Euclidean Gaussian likelihood is*

$$L_\sigma(b, \Delta_W) = \exp\left(-\frac{\|P_{W^\perp} b\|_2^2}{2\sigma^2}\right).$$

Proof. By Theorem 2.4, $\Delta_W b = P_{W^\perp} b$. Substitute this identity into the definition of $L_\sigma(b, A)$. □

Proposition 4.15 (Single-observation optimal classical guess)**.** *If $b \neq 0$ and $W_b = \mathrm{span}\{b\}$, then*

$$L_\sigma(b, \Delta_{W_b}) = 1.$$

Equivalently, the projection residual of b against the Laplacian whose kernel is $\mathrm{span}\{b\}$ is zero.

Proof. Since $b \in W_b$, $\Delta_{W_b} b = 0$. Hence $L_\sigma(b, \Delta_{W_b}) = \exp(0) = 1$. □

This proposition is deliberately local: it concerns one observation b and the candidate subspace generated by that same observation. It does not collapse the HCP search problem, because the hardness comparison ranges over the whole hidden Grassmannian family and over public observations that do not reveal the operative kernel.

Proposition 4.16 (Gaussian prior factorization)**.** *For $x = (x_i)_{i=1}^n \in R^n$,*

$$\exp\left(-\frac{\|x\|_2^2}{2}\right) = \prod_{i=1}^n \exp\left(-\frac{x_i^2}{2}\right).$$

Proof. Since $\|x\|_2^2 = \sum_i x_i^2$, the left hand side is

$$\exp\left(\sum_i -\frac{x_i^2}{2}\right),$$

which equals the product of the exponentials. □

Definition 4.17 (Spectral fingerprint)**.** *A finite-dimensional sheaf Laplacian is a self-adjoint positive semidefinite linear operator $L : V \to V$. Its spectral fingerprint is the multiset*

$$\mathrm{Spec}_{\mathrm{multi}}(L)$$

of eigenvalues counted with multiplicity. Two Laplacians are spectrally equivalent when these multisets agree. In contrast, the coarse homotopy equivalence surface used here records only harmonic dimension:

$$L_1 \simeq_{\ker} L_2 \quad \Longleftrightarrow \quad \dim\ker L_1 = \dim\ker L_2.$$

The order of determination is important. The public homotopy-class label α gives the public fiber A_α; it is not itself a fibration ID. A hidden representative $x \in A_\alpha$ determines the operative Hodge–Laplacian Δ_x. Once an HCP witness has selected or certified the hidden Hodge data (Δ_x, h, e), the fibration ID is the spectral fingerprint

$$\mathrm{FID}(x) := \mathrm{Spec}_{\mathrm{multi}}(\Delta_x).$$

Thus the spectral fingerprint gives the fibration identity in the certified Hodge layer, while remaining downstream of harmonic-cochain recovery and not part of the public homotopy class.

Proposition 4.18 (Spectrum is not determined by harmonic dimension)**.** *There exist self-adjoint positive semidefinite Laplacians L_1, L_2 with*

$$\dim\ker L_1 = \dim\ker L_2$$

but

$$\mathrm{Spec}_{\mathrm{multi}}(L_1) \neq \mathrm{Spec}_{\mathrm{multi}}(L_2).$$

Proof. Take $V = R$, $L_1 = \mathrm{id}_R$, and $L_2 = 2\,\mathrm{id}_R$. Both kernels are zero, so the harmonic dimensions agree. The spectral fingerprints are the singleton multisets $\{1\}$ and $\{2\}$, respectively, so the operators are not spectrally equivalent. □

Proposition 4.19 (Spectral-gap admissibility)**.** *If a candidate Hodge–Laplacian geometry carries a by-construction PoV/Tsirelson admissibility witness, then*

$$\lambda_1^+(\Delta) \in \left[\frac{\sqrt{2}}{2},\, 2\sqrt{2}\right].$$

Proof. The by-construction witness contains

$$\frac{\sqrt{2}}{2} \leq \lambda_1^+(\Delta) \qquad \text{and} \qquad \lambda_1^+(\Delta) \leq 2\sqrt{2}.$$

These two inequalities are exactly membership in the displayed closed interval. □

Thus the by-construction witness is an admissibility certificate for the displayed spectral window. It constrains which candidate objects are admissible; it does not remove hidden-operator ambiguity. The HCP hardness

mechanism in Section 2 remains the hidden fibration, hidden Laplacian, and hidden kernel.

For a nonempty family $\Omega \subseteq \mathcal{L}_1(R^n)$, the associated classical and quantum adversary-bound forms are spectral-statistical targets, not HCP hardness assumptions. The corresponding empty-family statements are only boundary checks; they are not nontrivial hardness bounds.

4.7 Heyting And Oracle Judgment Surface

The ownership split is:

$$\begin{aligned} \text{hidden equality} &: x \simeq y \iff \mathcal{P}(x,y) \neq \varnothing, \\ \text{public class data} &: x \equiv_\pi y \iff \pi(x) = \pi(y), \\ \text{public truth data} &: q : \Omega \longrightarrow \Omega_3. \end{aligned}$$

The first line is the Helfer path/diagonal interface, together with the usual sheaf-theoretic interpretation of intuitionistic logic [Hel19]. The second line is the public classifier interface defined above. The third line is the finite public judgment interface. We take

$$\Omega_3 = \{\bot, ?, \top\}, \qquad \bot < ? < \top,$$

with meet and join given by minimum and maximum in the chain. Its implication is the residuated operation characterized by

$$c \leq (a \Rightarrow b) \quad \iff \quad a \wedge c \leq b.$$

The quotient map q is not assumed faithful:

$$\exists\, \omega_1 \neq \omega_2 \quad \text{with} \quad q(\omega_1) = q(\omega_2).$$

The oracle-specific part is the query/answer mechanism that produces these public judgments. For this boundary we use oracle-modality terminology compatible with Ahman–Bauer [AB26]: oracle-visible conditions such as vertex compatibility, Betti-number compatibility, public identity-hash compatibility, and spectral compatibility are evaluated in the finite public Heyting algebra. Thus an oracle answer is a public quotient of hidden semantic evidence, not a reconstruction of the hidden representative, hidden path witness, fibration, Laplacian, or harmonic kernel.

Operational maps out of Ω_3, such as protocol actions attached to the intermediate public judgment ?, belong to the subsequent protocol paper's

runtime and consensus semantics. They are not the Heyting implication operation and do not alter the residuation law above.

The identity hash used by the oracle interface is an observation obtained from the public Hodge projection and quantization path. It is not a cochain commitment payload. The corresponding stability assumption says that equal public identity inputs give the same identity hash; it is a determinism condition for public identity observations, not a uniqueness or binding theorem for cochain commitments. Similarly, topological observations such as Betti-number compatibility are homotopy-invariant public judgments. Spectral fingerprints are post-recovery fibration-identification data: they are not determined by homotopy class alone by Theorem 4.18. The oracle interface therefore keeps Betti/topology observations, spectral admissibility, identity hashes, and cochain commitments as separate public conditions.

This modular split is part of the mathematical statement: cryptographic assumptions, public classifiers, and Hodge-theoretic claims are not identified. Exact and oracle invariants are used here as mathematical acceptance conditions; their full CS/protocol treatment is deferred to the subsequent protocol paper.

Lean Cross-Reference

The semantic and axiom interfaces in this section correspond to `Axioms.HelferFOL`, `Axioms.Fibration`, `Axioms.PublicClassifierFactorization`, `Axioms.PublicClassifierFactorization.publicFiber_eq_iUnion_baseFibers`, `Axioms.HomotopyClass`, `Axioms.Foundation`, `Axioms.FoundationWellFormed`, `Axioms.SheafCoherence`, `Axioms.SheafCoherenceWellFormed`, `Axioms.CommittedCochainWellFormed`, `Axioms.RepresentsExactOneCochain`, `Axioms.IsOneCocycle`, `Axioms.oneCocycle_of_committedCochainWellFormed`, `Gates.Exact`, `Gates.structural_preconditions_of_sheaf_exact_gates`, `Axioms.SpectralBounds`, `Tsirelson`, `Gates.Spectral`, `Axioms.StageAdmissibility`, `Gates.Stage`, `Axioms.Heyting`, and `Axioms.OracleModality`.

5 Cochain Commitments: Private Witnesses And Public Artifacts

This section treats cochain commitments at the boundary between private witness data and public-facing artifacts. In this section, "artifact" means public-facing artifact unless explicitly qualified. The commitment interface itself is mixed: commitment values, payloads, hashes, roots, and verification statements are public-facing, while openings, randomness, full cochains,

and hidden Hodge data remain witness-level unless a specified protocol path exposes them. These public-facing objects are the scaffolding for transparency in action and anonymity in identity: actions are exposed through auditable public artifacts, while identities remain inside hidden representatives and public equivalence classes. This section realizes the abstract public observation (α, b) of Theorem 2.6: the public homotopy-class label is α, while accepted cochain payloads, specified partial openings, and proof-carrying statements supply the cochain-facing datum b without presenting the hidden representative, Hodge–Laplacian, or harmonic kernel.

5.1 Cochain Commitments

Cochain commitment payloads provide the public handle on hidden cochain data. The cochain data itself lives at three different visibility levels. Fix a hidden representative $x \in A_\alpha$, with induced cochain complex $C_x^\bullet$, coboundaries δ_0, δ_1, and Hodge–Laplacian Δ_x.

(i) The private witness level contains the actual C_x^1-data: the harmonic component h, the noise component e, the full observed cochain $b = h+e$ when it is not explicitly published, the opening data (m, r), and the hidden restriction/Hodge data determining Δ_x.

(ii) The public payload level contains only a serialized residue edge vector

$$\bar{c} = \delta_0 a \pmod{q}, \qquad a = \mathrm{enc}(m, r) \in C_x^0,$$

together with a modulus, an edge count, and a topology hash. This is the cochain commitment payload. It is public as a committed C^1-level artifact, but it is not the private opening, not the full hidden cochain with Hodge decomposition, and not a presentation of Δ_x or $\ker(\Delta_x)$.

(iii) The opening/proof level may reveal local restrictions or certify equations involving $\bar{c}$. Only the stated local values, conditions, or proof statements become public.

Thus the cochain-facing observation b in Theorem 2.6 is the abstract datum supplied by an accepted payload, opening, or proof-carrying statement; the commitment payload records the public residue edge vector $\bar{c}$; and the hidden cochain witness remains private unless a protocol path explicitly exposes a specified part of it. The canonical payload is a mathematical object with a modulus, public residue edge values, declared edge count, and topology hash.

Definition 5.1 (Cochain commitment payload)**.** *A cochain commitment payload is a tuple*

$$(q, n, c, \tau)$$

consisting of a modulus q, a declared edge count n, a list c of integer edge values, and a topology hash τ.

The intended commitment map is the transaction-facing coboundary map

$$(m, r) \longmapsto \delta(m\|r) \in C^1 \pmod q,$$

stored together with the topology hash of the supporting complex. The payload therefore preserves the accepted modulus, the edge-vector shape, the public topology binding, and the serialized C^1-level commitment values. It does not expose the hidden fibration representative, private transport data, the operative Hodge–Laplacian, or the harmonic kernel.

Definition 5.2 (Commitment well-formedness)**.** *A cochain commitment payload is well formed when its modulus is the supported canonical modulus $q = 12289$, its declared edge count agrees with the actual edge-value list length, and the declared edge count is positive.*

This payload predicate is deliberately only a wire-shape condition. Relative to an ambient sheaf S, the stronger mathematical acceptance condition is the committed one-cochain coherence predicate $c \in \mathcal{G}_S$ from Theorem 4.5: it additionally requires coherent typed coboundaries, binding to S, and a witness $\mathrm{dec}_S(c) = \delta_0 a$. Consequently an accepted committed one-cochain is a cocycle by Theorem 4.6.

Proposition 5.3 (Well-formed committed payloads)**.** *A well-formed cochain commitment payload has nonempty edge data and modulus greater than one.*

Proof. By well-formedness, the declared edge count equals the edge-list length and is positive, so the edge list is nonempty. The modulus is $12289 > 1$. □

Proposition 5.4 (Topology-compatible aggregation)**.** *Commitment aggregation is only meaningful for payloads with compatible modulus, declared edge count, and topology hash. This is the public topology-binding condition consumed by the protocol layer.*

Proof. This is the definition of aggregation compatibility: both payloads must be well formed, and their modulus, declared edge count, and topology hash must agree. Without these equalities the payload-level addition operation is not an accepted aggregation. □

The security interface for this commitment layer consists of binding, hiding, homomorphic addition, homomorphic scalar multiplication, topology binding, and partial-opening soundness. The blinded-commitment, vector-commitment, and cohomological-MAC interfaces are separate axiom records rather than consequences silently bundled into the payload shape.

Definition 5.5 (Blinded cochain commitment boundary)**.** *Fix an accepted exact payload c, with decoded value $\mathrm{dec}_S(c) = \delta_0 a$, and write s for a separate harmonic blinding scalar. A typed Hodge-blinding boundary consists of the following data and conditions:*

(i) C^1 *has the orthogonal Hodge splitting*

$$C^1 = \mathcal{H}^1 \oplus \mathrm{im}(\delta_0) \oplus \mathrm{im}(\delta_1^*), \qquad \mathcal{H}^1 \perp \mathrm{im}(\delta_0) \perp \mathrm{im}(\delta_1^*).$$

(ii) For each public topology hash τ, the blinding direction $H_\tau \in \mathcal{H}^1$ is determined by τ.

(iii) The associated blinded value has the form

$$c_\tau^{\mathrm{blind}}(a, s) = \mathrm{dec}_S(c) + sH_\tau = \delta_0 a + sH_\tau \in C_S^1.$$

(iv) The exact component $\delta_0 a$ is orthogonal to H_τ.

(v) Binding and hiding for the resulting commitment scheme are separate cryptographic assumptions. In particular, exactness and Hodge orthogonality supply the geometric decomposition but do not by themselves prove a distributional hiding game.

The distinction between a and s is intentional. The former is the degree-zero opening whose coboundary gives the accepted public payload; the latter scales the topology-derived harmonic direction. Moreover, the serialized exact payload and the associated Hodge-blinded value are related objects, not definitionally the same wire object. In particular, the modular wire representation is interpreted by dec_S before the Hodge-space addition is formed; the paper does not identify a real cochain space with the quotient C^1/qC^1.

Definition 5.6 (Vector commitment boundary)**.** *A vector commitment boundary is well formed if and only if all of the following hold:*

(i) For vertex values $a = (a_v)$, the committed edge cochain is $c = \delta a$, i.e. on an oriented edge $e = (u, v)$,

$$c_e = (\delta a)_e = R_{v\to e} a_v - R_{u\to e} a_u.$$

(ii) An opening over a revealed set S is a pair

$$o_S = (a'_S, \omega_S),$$

where $a'_S = (a'_v)_{v \in S}$ is the claimed restriction and ω_S is its authentication witness.

(iii) Completeness: for every revealed set S of vertices, the honest opening

$$o_S^{\mathrm{hon}} = (a|_S, \mathrm{Open}_S(c))$$

satisfies $\mathrm{Ver}_\tau(c, S, o_S^{\mathrm{hon}}) = 1$.

(iv) Soundness: if $o_S = (a'_S, \omega_S)$ and

$$\mathrm{Ver}_\tau(c, S, o_S) = 1,$$

then every revealed incidence satisfies the same coboundary equation $c_e = R_{v \to e} a'_v - R_{u \to e} a'_u$.

(v) Position binding: the public topology hash fixes the vertex and edge positions with respect to which openings are interpreted.

Definition 5.7 (Cohomological MAC boundary)**.** *A cohomological MAC boundary is well formed if and only if all of the following hold:*

(i) The key determines a Hodge–Laplacian Δ and harmonic projection $P_H : C^1 \to \ker(\Delta)$.

(ii) Correct tags have the form

$$T_{\Delta,\nu}(m) = P_H c(m, \nu)$$

for the committed cochain $c(m, \nu)$ determined by the message and nonce.

(iii) Verification is the Hodge-projection check

$$\mathrm{Ver}_\Delta(m, \nu, T) = 1 \iff \Delta T = 0 \quad \textit{and} \quad P_H T = P_H c(m, \nu).$$

(iv) Correctness:

$$\mathrm{Ver}_\Delta(m, \nu, T_{\Delta,\nu}(m)) = 1.$$

(v) Unforgeability: producing a verifying tag without the key is exactly the stated hidden-Hodge projection condition for this interface.

(vi) Key binding: accepted tags are bound to the chosen Hodge–Laplacian key.

(vii) Determinism: the same key, message, and nonce determine the same tag.

Full adversarial games for hiding, unforgeability, and partial openings belong to the subsequent protocol paper.

The commitment-facing invariants attach to the commitment payload and, after packaging, to the public artifact layer; they are not proximity projection data. Payload well-formedness is a shape condition: it asserts the supported modulus, agreement between the declared edge count and the edge-value list, and a nonempty edge list. It is not a cryptographic uniqueness theorem and it does not assert that the cochain vector is nonzero. For example, the canonical zero payload is well formed whenever the declared edge count is positive.

Topology compatibility is a separate public invariant. Two committed cochains can be aggregated only when their moduli, declared edge counts, and topology hashes agree. The weaker same-topology relation records equality of topology hashes alone. Cryptographic binding and topology-binding claims are supplied by the commitment-security axioms, not by the raw payload data.

At the artifact level, a hashed committed cochain is a public cochain payload together with digest data and a binding assertion. A proof-carrying committed cochain adds a proof system, a statement, and a verification condition. The well-formedness predicate for the proof-carrying artifact includes well-formedness of the hashed public artifact; well-formedness of that hashed artifact in turn includes well-formedness of its underlying cochain payload. The basic theorem exposed to this paper is that a well-formed proof-carrying committed-cochain artifact has an underlying well-formed cochain commitment payload.

When a sheaf context S is available, the refined artifact predicate requires the underlying payload to satisfy $c \in \mathcal{G}_S$, not merely payload well-formedness. This refinement is inherited by the corresponding proof-carrying and runtime-acceptance predicates. It separates two valid but different claims: ordinary artifact well-formedness exposes a valid serialized payload, whereas sheaf-coherent artifact well-formedness exposes an exact committed one-cochain and therefore a one-cocycle.

Proposition 5.8 (Proof-carrying artifact exposes payload)**.** *A well-formed proof-carrying committed-cochain artifact exposes a well-formed underlying*

cochain commitment payload.

Proof. Unpack proof-carrying well-formedness to obtain well-formedness of the hashed public artifact, then unpack hashed-artifact well-formedness to obtain well-formedness of its cochain payload. □

Proposition 5.9 (Sheaf-coherent proof artifact exposes an exact cocycle)**.** *If a proof-carrying committed-cochain artifact is coherent with S, then its underlying payload c lies in $\mathcal{G}_S$ and satisfies*

$$\delta_1 \operatorname{dec}_S(c) = 0.$$

Proof. The refined artifact predicate projects to $c \in \mathcal{G}_S$. Apply Theorem 4.6. □

5.2 Public Artifacts

The public-facing stack has three mathematical levels. The fully public handle level contains hashes, roots, topology hashes, nullifiers, public summaries, and declared public parameters. The verifier-facing artifact level contains compressed public data, cochain commitment artifacts, proof-carrying cochain payloads, and certificate data. The private witness level contains the fibration representative, transport/restriction data, full cochains and openings, and the Hodge operator whose kernel selects the harmonic component. The paper's claims are stated so that public handles and verifier-facing artifacts bind to hidden witnesses without turning the hidden witness into public data. This paper records the mathematical public/private separation required by the sheaf-theoretic attacker-observation viewpoint; the associated security games and runtime observation model are treated in the subsequent protocol paper.

These layers form the following public artifact stack:

hidden/private object
→ compressed public datum
→ commitment/root/hash artifact
→ proof-carrying artifact.

Verifier and oracle judgments are conditions evaluated on the terminal public artifact, not further objects in the stack. Each layer binds to or summarizes the preceding layer. A hash does not replace the mathematical object, a commitment does not replace the witness, and a proof artifact does not

replace the proof statement. The purpose of the stack is to let public verification refer to hidden sheaf/cochain data without exposing it. In this sense, the stack separates what must be publicly accountable from what must remain representative-private.

Hashes serve as public names, integrity bindings, and fixed-size pointers for larger artifacts. A topology hash names the public combinatorial support; a root commits to a batch or storage view; a transcript hash domain-separates a proof statement. None of these hashes asserts validity by itself. Validity comes from the condition or proof attached to the hashed object.

5.3 Lossy Projection And Preserved Structure

Storage roots, topology hashes, cochain roots, nullifiers, and proof objects are public artifacts in the stack. Full sheaf, cochain, and fibration data remain witness material unless a protocol path explicitly exposes them. The storage mechanics are protocol-level concerns; the mathematical point here is that stored public artifacts are compressed images of accepted witness data, not replacements for the hidden object.

In the notation of Theorem 2.6, α is the public homotopy-class label obtained as a lossy classifier value. Its fiber is

$$A_\alpha = \pi^{-1}(\alpha),$$

the set of hidden representatives compatible with the same public class. For $x \in A_\alpha$, the induced cochain complex $C_x^\bullet$, Hodge–Laplacian Δ_x, and harmonic kernel $\ker(\Delta_x)$ are not determined by α.

Compression creates verifier-facing public artifacts from hidden or full witness data. The compression need not be injective; it must preserve enough structure to support the relevant public statement.

A compressed public datum is allowed to forget hidden data. In fact, it must forget enough data that it does not select a private representative. The relevant question is not whether the public datum is injective, but which facts survive the compression relation.

The preserved structure before the finite-code layer is:

(i) public homotopy-class equality for hidden representatives;

(ii) cochain commitment payload well-formedness;

(iii) topology binding and exact-one-cochain realization for accepted sheaf-coherent commitments;

(iv) digest/root binding for compressed public data;

(v) proof-carrying validity of the public statement attached to an artifact.

The lost structure is the private representative itself: full sheaf/cochain witness data, hidden transport, the operative Hodge operator, private randomness, and runtime lifecycle facts. Those may be bound to public artifacts or certified by a proof-carrying statement, but they are not made public merely by the presence of a hash, root, or commitment payload.

Byte parsing, allocation bounds, canonical encoding, concrete hash functions, domain separation, storage inclusion proofs, proof verification, state updates, and ephemeral-fiber lifecycle properties are TCB requirements. They are boundary conditions, not part of the mathematical cochain commitment payload.

For HCP, the concrete sources for the abstract observation (α, b) are: α is the public homotopy-class label, and b is the cochain-facing datum supplied by an accepted cochain payload, opening, or proof-carrying statement. For $x \in A_\alpha$, private restriction data determines $C_x^\bullet$, δ_0, δ_1, Δ_x, and the recovery target

$$h = \pi_{H_x}(b) \in \ker(\Delta_x),$$

when $b \in C_x^1$.

The artifacts in this section may bind to the hidden representative or certify equations involving b, but they do not present Δ_x, $\ker(\Delta_x)$, or the private fibration data. Once HCP has selected or certified the hidden Hodge data, the fibration ID is

$$\mathrm{FID}(x) = \mathrm{Spec}_{\mathrm{multi}}(\Delta_x).$$

5.4 Plucker Coordinates

Let V be a finite-dimensional vector space over a field K, and let $1 \le r \le \dim V$. A private rank-r frame

$$\mathcal{F} = (w_1, \ldots, w_r)$$

spans an r-plane

$$W = \langle w_1, \ldots, w_r \rangle \in \mathrm{Gr}(r, V).$$

The Plucker embedding sends this Grassmannian point to

$$\mathrm{Pl}(W) = [w_1 \wedge \cdots \wedge w_r] \in \mathbb{P}(\bigwedge^r V).$$

A Plucker coordinate is the public geometric datum obtained from such a private frame: a representative of $\mathrm{Pl}(W)$, or a digest-bound canonical encoding of such a representative, equipped with the relevant projective well-formedness conditions. It is required to respect projective rescaling and the Plucker relations; changing the private frame for the same Grassmannian point does not change the normalized Plucker coordinate. This is the Plucker coordinate used by the proximity projection, not the HCP public class label α fixed above.

Thus the frame $\mathcal{F}$ is witness material, while the Plucker coordinate is the verifier-facing projective datum. It is not itself the witness. It can be bound by a hash, root, or proof-carrying statement, but the actual witness remains the hidden frame or representative.

In the proximity specialization of Section 6, the cochain commitment remains the public committed payload. The Plucker coordinate enters as witness geometry for a public-binding statement

$$\big(\mathrm{Pl}(\mathcal{F}), c\big) \in \mathcal{R}.$$

For the functional encoder, the projection direction is

$$\mathrm{Pl}(\mathcal{F}) \xrightarrow{E} \underbrace{c}_{\text{accepted cochain commitment}} \longrightarrow \text{plane/RS/MCA verifier data.}$$

A proof-carrying binding relation asserts that the accepted cochain commitment realizes the Plucker coordinate.

Lean Cross-Reference

The public artifact and commitment interfaces in this section correspond to the following Lean declarations:

- Plucker coordinates and public binding: `Proximity.GrassmannianPlucker`, `Proximity . GrassmannianPlucker . Concrete`, and `Proximity . PluckerCochain.PublicBinding`;
- payload shape and invariants: `Axioms . CochainCommitmentData`, `CochainCommitmentData . WellFormed`, `CochainCommitmentData . AddCompatible`, `CochainCommitmentData . edge _ values _ nonempty _ of _ wellFormed`, and `CochainCommitmentData . modulus _ gt _ one _ of _ wellFormed`;
- sheaf-semantic commitment invariants: `Axioms . CommittedCochainWellFormed`, `CochainCommitmentData . CoherentWith`, `Axioms . RepresentsExactOneCochain`, `Axioms . IsOneCocycle`, and `Axioms.oneCocycle_of_committedCochainWellFormed`;

- commitment-security interfaces: `Axioms . CochainCommitmentAxioms`, `Axioms . HodgeBlindingContext`, `Axioms . BlindedCommittedCochain`, `Axioms . BlindedCommittedCochain . base _ representsExactOneCochain`, `Axioms . BlindedCommittedCochain . base _ isOneCocycle`, `Axioms . BlindedCommittedCochain . harmonicDirection _ isHarmonic`, `Axioms . BlindedCommittedCochain . exactComponent _ orthogonal _ harmonicDirection`, `Axioms . BlindedCommittedCochain . exists _ exact _ harmonic _ decomposition`, `Axioms . BlindedCommitmentAxioms`, `Axioms . BlindedCommitmentWellFormed`, `Axioms . VectorCommitmentAxioms`, `Axioms . VectorCommitmentWellFormed`, `Axioms . CohomologicalMACAxioms`, and `Axioms . CohomologicalMACWellFormed`;
- proof-carrying artifacts: `TCB . HashedCommittedCochain`, `TCB . ProofCarryingCommittedCochain`, and `TCB . proofCarrying _ cochain _ wellFormed`;
- sheaf-coherent artifact refinements: `TCB . HashedCommittedCochain . CoherentWith`, `TCB . ProofCarryingCommittedCochain . CoherentWith`, `TCB . proofCarrying _ cochain _ coherentWith`, and `TCB . RuntimeAcceptsSheafCommittedCochain`.

6 Reed–Solomon Projection And Proximity

The staged projection maps are

$$\mathrm{Pl}(\mathcal{F}) \xrightarrow{E} c \xrightarrow{\mathrm{proj}_{\mathcal{E}}} p_c \xrightarrow{\Phi_*} y.$$

Here c is the chain's cochain payload, p_c is its plane shadow, and y is the finite-field received word. The public-binding relation introduced below identifies c with an accepted cochain commitment. For a valid frame, the corresponding stage predicates are

$$\sigma(c) \in Z, \qquad p_c \in \mathcal{C}_{\mathcal{P}}, \qquad y \in T.$$

Thus the local syndrome and canonical Reed–Solomon target are acceptance conditions on projected objects, not additional projection maps. The chain has two complementary proof directions. The stage maps give codeword preservation and the near-side metric handoff

$$D(y) \le d_L(c) \le \rho(\mathcal{F}),$$

while a supplied final far-region theorem gives the calibrated non-collapse implication needed for a gap reduction. Once both directions have been proved, they yield the end-to-end proximity statement. The projective

Plucker coordinate is the geometric handoff: a private frame determines $\mathrm{Pl}(\mathcal{F})$, and the encoder E produces the chain cochain c. At the public boundary, the relational form $\mathcal{R}$ identifies that payload with an accepted cochain commitment.

At the projection-chain level, the private source datum is the frame witness $\mathcal{F}$. When this chain is integrated with the semantic and commitment layers of Sections 4 and 5, that frame may be accompanied by fibration data, paths, and commitment openings. The verifier-facing mathematical data then consist of the accepted cochain commitment, the Plucker coordinate bound to it, the plane projection, and the finite received word. Hashes and roots are compressed handles for canonical bytes of those data, not replacements for the mathematical objects.

6.1 The End-to-End Projection Contract

The displayed chain must do more than produce a finite word. On its near side, it must carry every valid frame to a Reed–Solomon codeword and control the target distance by the frame disagreement:

$$\mathcal{F} \in \mathcal{V}_{\mathrm{fr}} \Longrightarrow y \in T, \qquad D(y) \le \rho(\mathcal{F}).$$

On its far side, it must not collapse a source instance beyond the calibrated target radius. The stage maps prove the first two statements; a separate far-region witness must prove the non-collapse statement. These are precisely the two directions required to pull a target gap solver back through the composed map.

We now isolate that contract in notation that will be instantiated by the concrete stages below. A source proximity datum is the three-field record

$$\mathcal{X} = (X, G_X, \rho_X),$$

where X is the source carrier, $G_X \subseteq X$ is its exact or valid locus, and $\rho_X : X \to \mathbb{R}$ is its disagreement score. End to end, these are respectively the frame-witness carrier, the valid-frame locus, and the Plucker disagreement. At an intermediate stage the same interface may be instantiated by committed cochains and their local syndrome distance. A code-side proximity datum is the three-field record

$$\mathcal{Y} = (Y, \mathcal{C}_Y, d_Y),$$

where Y is the target word carrier, $\mathcal{C}_Y \subseteq Y$ is the code, and $d_Y : Y \to \mathbb{R}$ is the distance-to-code score.

Definition 6.1 (Evaluation projection)**.** *An evaluation projection $\eta : \mathcal{X} \to \mathcal{Y}$ is a map*

$$\eta : X \longrightarrow Y$$

such that

$$x \in G_X \Longrightarrow \eta(x) \in \mathcal{C}_Y, \qquad d_Y(\eta(x)) \leq \rho_X(x) \quad (x \in X).$$

Consequently, if $\rho_X(x) \leq \varepsilon$, then

$$d_Y(\eta(x)) \leq \varepsilon.$$

A promised gap-proximity problem is a tuple

$$P = (X, s, r_{\text{near}}, r_{\text{far}}) \qquad \text{with} \qquad r_{\text{near}} \leq r_{\text{far}}.$$

A promised instance has near and far loci

$$P_{\text{near}} = \{x \in X : s(x) \leq r_{\text{near}}\}, \qquad P_{\text{far}} = \{x \in X : r_{\text{far}} \leq s(x)\}.$$

A solver $A : X \to \{\text{accept}, \text{reject}\}$ is correct on the promise when

$$x \in P_{\text{near}} \Rightarrow A(x) = \text{accept}, \qquad x \in P_{\text{far}} \Rightarrow A(x) = \text{reject}\,.$$

A gap reduction $f : P \to Q$ preserves near instances and far instances. Hence any correct solver for Q pulls back to a correct solver for P.

Proposition 6.2 (Projection and solver pullback)**.** *Let $\eta : \mathcal{X} \to \mathcal{Y}$ be an evaluation projection. Then:*

$$\rho_X(x) \leq \varepsilon \Longrightarrow d_Y(\eta(x)) \leq \varepsilon.$$

If, in addition, the projection is calibrated by radii

$$r^X_{\text{near}} \leq r^X_{\text{far}}, \qquad r^Y_{\text{near}} \leq r^Y_{\text{far}}, \qquad r^X_{\text{near}} \leq r^Y_{\text{near}},$$

and satisfies

$$r^X_{\text{far}} \leq \rho_X(x) \Longrightarrow r^Y_{\text{far}} \leq d_Y(\eta(x)),$$

then η is a gap reduction from the source gap problem to the target gap problem. Consequently, every correct target solver A pulls back to the correct source solver $A \circ \eta$.

Proof. The first assertion is transitivity:

$$d_Y(\eta(x)) \le \rho_X(x) \le \varepsilon.$$

For the calibrated near side, if x is source-near, then

$$d_Y(\eta(x)) \le \rho_X(x) \le r^X_{\text{near}} \le r^Y_{\text{near}},$$

so $\eta(x)$ is target-near. The far side is exactly the displayed non-collapse implication. If A is correct on the target promise, then for a source-near x, the just-proved near preservation gives

$$A(\eta(x)) = \text{accept},$$

and for a source-far x, the far preservation gives

$$A(\eta(x)) = \text{reject}\,.$$

This is precisely correctness of the pulled-back solver. □

For a lossy projection chain, the final interface is calibrated. A calibrated end-to-end gap projection consists of an evaluation projection $\eta : \mathcal{X} \to \mathcal{Y}$, radii

$$r^X_{\text{near}} \le r^X_{\text{far}}, \qquad r^Y_{\text{near}} \le r^Y_{\text{far}},$$

the near compatibility

$$r^X_{\text{near}} \le r^Y_{\text{near}},$$

and the far non-collapse implication

$$r^X_{\text{far}} \le \rho_X(x) \Longrightarrow r^Y_{\text{far}} \le d_Y(\eta(x)).$$

The ordinary same-radius projection is the special case $r^X_{\text{near}} = r^Y_{\text{near}}$ and $r^X_{\text{far}} = r^Y_{\text{far}}$.

6.2 Grassmannian Source And Plucker Quotient

The first Grassmannian context scores an ambient observation against one fixed harmonic subspace. Let $W \le \mathbb{R}^n$ be a Grassmannian point with orthogonal projection P_W, and put

$$\Delta_W = I - P_W.$$

Its carrier is $\mathbb{R}^n$, the exact locus is

$$G_W = W,$$

and the disagreement score is

$$\rho_W(b) = \|P_{W^\perp} b\|.$$

The residual identity is

$$\rho_W(b) = \mathrm{res}_W(b),$$

and

$$\rho_W(b) = 0 \iff b \in W \iff b \in \ker(\Delta_W).$$

The identity map is therefore an evaluation projection from this vector-level Grassmannian proximity datum to its harmonic code shadow.

Proposition 6.3 (Grassmannian residual)**.** *For $W \leq \mathbb{R}^n$ with orthogonal projection P_W,*

$$\rho_W(b) = \|P_{W^\perp} b\| = \mathrm{res}_W(b),$$

and

$$\rho_W(b) = 0 \iff b \in W \iff b \in \ker(\Delta_W).$$

Proof. The projection model has

$$b = P_W b + P_{W^\perp} b, \qquad \Delta_W b = (I - P_W)b = P_{W^\perp} b.$$

Thus the Laplacian residual is

$$\mathrm{res}_W(b) = \|\Delta_W b\| = \|P_{W^\perp} b\| = \rho_W(b).$$

Since a norm vanishes exactly on the zero vector,

$$\rho_W(b) = 0 \iff P_{W^\perp} b = 0.$$

The latter condition is equivalent to $b = P_W b$, hence to $b \in W$. In the projection Hodge model, harmonicity is membership in $\ker(\Delta_W) = W$. □

The composed chain below uses a second, downstream Grassmannian context: its source is a frame witness and its disagreement $\rho(\mathcal{F})$ is supplied with that frame-level proximity instance. The vector residual $\rho_W(b)$ and the frame score $\rho(\mathcal{F})$ are therefore not identified merely by sharing the same underlying harmonic subspace; an application relating them must state the comparison explicitly.

Now fix a rank r Grassmannian point. A frame witness is

$$\mathcal{F} = (W; w_1, \ldots, w_r)$$

with $w_i \in W$, linearly independent, and

$$\operatorname{span}_{\mathbb{R}}\{w_1, \dots, w_r\} = W.$$

Its exterior coordinate is

$$x_{\mathcal{F}} = w_1 \wedge \cdots \wedge w_r \in \bigwedge^r \mathbb{R}^n.$$

The raw Plucker validity conditions are

$$x_{\mathcal{F}} \neq 0, \qquad x_{\mathcal{F}} \text{ is decomposable}, \qquad \mathsf{PluckerRelations}(x_{\mathcal{F}}).$$

Here $\mathsf{PluckerRelations}$ is the relation selected by the framed system; the present unrestricted projective instantiation takes it to be decomposability. Public-shadow validity additionally requires normalization and support admissibility. Let $\mathcal{V}_{\mathrm{fr}}$ denote the locus of frame witnesses satisfying both the raw-coordinate and public-shadow validity conditions. In the concrete projective instantiation,

$$x \sim y \iff \exists u \in \mathbb{R}^{\times},\ y = ux,$$

and the Plucker coordinate is the projective class

$$\mathrm{Pl}(\mathcal{F}) = [x_{\mathcal{F}}] \in \mathbb{P}\left(\bigwedge^r \mathbb{R}^n\right).$$

The soundness conditions are:

$$\begin{gathered} x_{\mathcal{F}} \neq 0, \qquad \mathsf{PluckerRelations}(x_{\mathcal{F}}), \\ \mathrm{Pl}(\mathcal{F}) \text{ is normalized/support-admissible}, \\ \mathcal{F}.\mathsf{point} = \mathcal{G}.\mathsf{point} \Longrightarrow x_{\mathcal{F}} \sim x_{\mathcal{G}}, \qquad x \sim y \Longrightarrow [x] = [y]. \end{gathered}$$

Thus changing the private frame for the same Grassmannian point does not change the projective Plucker coordinate.

Proposition 6.4 (Projective Plucker invariance)**.** *Assume the displayed Plucker soundness conditions. If two frame witnesses $\mathcal{F} = (W; w_1, \dots, w_r)$ and $\mathcal{G} = (W; v_1, \dots, v_r)$ present the same Grassmannian point, then*

$$[w_1 \wedge \cdots \wedge w_r] = [v_1 \wedge \cdots \wedge v_r] \in \mathbb{P}\left(\bigwedge^r \mathbb{R}^n\right).$$

Proof. Let $x_{\mathcal{F}} = w_1 \wedge \cdots \wedge w_r$ and $x_{\mathcal{G}} = v_1 \wedge \cdots \wedge v_r$. The soundness condition for frame change gives $x_{\mathcal{F}} \sim x_{\mathcal{G}}$. By definition of the projective quotient, this means that for some $u \in \mathbb{R}^{\times}$,

$$x_{\mathcal{G}} = u x_{\mathcal{F}}.$$

The quotient map to projective space identifies nonzero scalar multiples; hence $[x_{\mathcal{F}}] = [x_{\mathcal{G}}]$. □

The source metric gate states

$$0 \le \rho(\mathcal{F}), \qquad \mathrm{Pl}(\mathcal{F}) = \mathrm{Pl}(\mathcal{G}) \Longrightarrow \rho(\mathcal{F}) = \rho(\mathcal{G}).$$

The Plucker metric gate fixes the admissible projective chart and a projective distance d_{Pl} satisfying

$$0 \le d_{\mathrm{Pl}}(\mathcal{F}, \mathcal{G}), \qquad \mathrm{Pl}(\mathcal{F}) = \mathrm{Pl}(\mathcal{F}') \Longrightarrow d_{\mathrm{Pl}}(\mathcal{F}, \mathcal{G}) = d_{\mathrm{Pl}}(\mathcal{F}', \mathcal{G}).$$

6.3 Committed Cochains And Local Syndrome

The committed-cochain carrier in the proximity bridge is exactly the cochain commitment payload of Section 5. Write

$$\mathcal{C}_{\mathrm{com}}$$

for that carrier. An element is the payload

$$c = (q, n, \mathbf{c}, \tau)$$

with supported modulus, declared edge count, edge values, and topology hash. Let

$$\mathcal{C}_{\mathrm{com}}^{\mathrm{wf}} \subseteq \mathcal{C}_{\mathrm{com}}, \qquad \mathcal{S}_{\mathrm{coh}}$$

denote respectively the well-formed commitment payloads and the coherent sheaf data. For $S \in \mathcal{S}_{\mathrm{coh}}$, define

$$G_S = \{c \in \mathcal{C}_{\mathrm{com}} : c \in \mathcal{C}_{\mathrm{com}}^{\mathrm{wf}}, \quad c \,\mathsf{binds}\, S, \quad \exists a \in C_S^0,\ \mathrm{dec}_S(c) = \delta_0 a\}.$$

Equivalently, for $S \in \mathcal{S}_{\mathrm{coh}}$, membership $c \in G_S$ is precisely the committed one-cochain coherence contract. Consequently,

$$c \in G_S \Longrightarrow c \in \mathcal{C}_{\mathrm{com}}^{\mathrm{wf}},$$

and the sheaf coherence hypotheses give

$$\delta_1 \delta_0 = 0, \qquad \Delta \text{ is built from the adjacent coboundaries.}$$

Moreover, exactness now applies to the particular payload:

$$\delta_1 \operatorname{dec}_S(c) = 0.$$

Thus G_S is the accepted committed-cocycle locus. It is not the degree-zero global-section locus. Its elements also have nonempty edge data and a supported modulus $q > 1$.

Definition 6.5 (Local syndrome target)**.** *Fix sheaf coherence data S. A local syndrome target is a quadruple*

$$\mathcal{L} = (\Sigma, \sigma, Z, d_L)$$

where

$$\sigma : \mathcal{C}_{\mathrm{com}} \to \Sigma, \qquad Z \subseteq \Sigma, \qquad d_L : \mathcal{C}_{\mathrm{com}} \to \mathbb{R},$$

and

$$c \in G_S \Longrightarrow \sigma(c) \in Z \qquad (c \in \mathcal{C}_{\mathrm{com}}).$$

It determines the local code

$$\mathcal{C}_L \subseteq \mathcal{C}_{\mathrm{com}}, \qquad c \in \mathcal{C}_L \Longleftrightarrow c \in G_S \wedge \sigma(c) \in Z.$$

The induced code-side proximity space has word carrier $\mathcal{C}_{\mathrm{com}}$, code $\mathcal{C}_L$, and distance d_L. Hence every accepted committed cocycle lies in the local code.

Definition 6.6 (Plucker-to-cochain projection)**.** *Let P_{Pl} be the projective Plucker quotient. A Plucker-to-cochain projection is a map*

$$E : P_{\mathrm{Pl}} \longrightarrow \mathcal{C}_{\mathrm{com}}$$

such that, for every frame witness $\mathcal{F}$,

$$\mathcal{F} \in \mathcal{V}_{\mathrm{fr}} \Longrightarrow E(\mathrm{Pl}(\mathcal{F})) \in G_S$$

and

$$d_L(E(\mathrm{Pl}(\mathcal{F}))) \leq \rho(\mathcal{F}).$$

A concrete payload encoder is a special case of this relation. It provides

$$E : P_{\mathrm{Pl}} \to \mathcal{C}_{\mathrm{com}}, \qquad N_{\mathrm{edge}} \in \mathbb{N}, \quad N_{\mathrm{edge}} > 0,$$

and, for valid frames,

$$E(\mathrm{Pl}(\mathcal{F})) \in G_S, \qquad n_{\mathrm{edge}}(E(\mathrm{Pl}(\mathcal{F}))) = N_{\mathrm{edge}}.$$

Thus the encoded edge vector has length N_{edge}. Since the encoder consumes the projective coordinate, equality of Plucker coordinates gives equality of the encoded committed cochain.

The storage-native statement is relational. A payload realization is a relation

$$\mathcal{R} \subseteq P_{\mathrm{Pl}} \times \mathcal{C}_{\mathrm{com}}$$

such that

$$(\xi, c) \in \mathcal{R} \Longrightarrow c \in G_S, \qquad \xi = \xi' \text{ and } (\xi, c) \in \mathcal{R} \Longrightarrow (\xi', c) \in \mathcal{R}.$$

An observed frame binding consists of a hashed Plucker coordinate, an observed proof-carrying cochain artifact, a valid frame, and a public statement proving

$$(\mathrm{Pl}(\mathcal{F}), c) \in \mathcal{R}.$$

It yields

$$c \in \mathcal{C}^{\mathrm{wf}}_{\mathrm{com}}, \qquad c \in G_S, \qquad c \in \mathcal{C}_L.$$

Proposition 6.7 (Observed Plucker binding gives a local codeword)**.** *Let an observed frame binding relative to S prove*

$$(\mathrm{Pl}(\mathcal{F}), c) \in \mathcal{R}$$

for $\mathcal{F} \in \mathcal{V}_{\mathrm{fr}}$. Then c is well formed, c is an accepted committed cocycle, and c is a codeword for every local syndrome target $\mathcal{L}$ over S.

Proof. The payload-realization relation includes the implication

$$(\xi, c) \in \mathcal{R} \Longrightarrow c \in G_S.$$

Applying it to $\xi = \mathrm{Pl}(\mathcal{F})$ gives $c \in G_S$, which in particular gives payload well-formedness and the degree-one cocycle equation. For a local syndrome target $\mathcal{L} = (\Sigma, \sigma, Z, d_L)$, the target axiom gives

$$c \in G_S \Longrightarrow \sigma(c) \in Z.$$

Together with $c \in G_S$, this is exactly $c \in \mathcal{C}_L$. □

Definition 6.8 (Syndrome adequacy gate)**.** *For a local syndrome target $\mathcal{L} = (\Sigma, \sigma, Z, d_L)$, the syndrome-adequacy gate is*

$$\mathcal{A}_S(\mathcal{L}) \Longleftrightarrow \forall c \in \mathcal{C}_{\mathrm{com}}, \quad c \in \mathcal{C}^{\mathrm{wf}}_{\mathrm{com}} \wedge (\sigma(c) \in Z) \Longrightarrow c \in G_S.$$

Proposition 6.9 (An adequate syndrome gives a local codeword)**.** *If* $\mathcal{A}_S(\mathcal{L})$*,* $c \in \mathcal{C}^{\mathrm{wf}}_{\mathrm{com}}$*, and* $\sigma(c) \in Z$*, then*

$$c \in \mathcal{C}_L.$$

Proof. The adequacy implication gives $c \in G_S$. Together with $\sigma(c) \in Z$, this is the defining conjunction for $c \in \mathcal{C}_L$. □

In the concrete commitment-sum bridge, the construction input is

$$c \in G_S \Longrightarrow \mathrm{sum}_{\mathrm{com}}(c) = 0.$$

The induced local syndrome target is

$$\Sigma = \mathbb{Z}, \qquad \sigma(c) = \mathrm{sum}_{\mathrm{com}}(c), \qquad Z = \{0\},$$

with binary residual

$$d_L(c) = \begin{cases} 0, & \mathrm{sum}_{\mathrm{com}}(c) = 0, \\ 1, & \mathrm{sum}_{\mathrm{com}}(c) \neq 0. \end{cases}$$

Global candidates have zero commitment-sum residual.

6.4 Plane Extraction And RS Encoding

Let I be the finite evaluation-index type. A two-coordinate plane word is

$$p : I \times \mathrm{Fin}(2) \to R.$$

A plane extraction context is

$$\mathcal{E} = (N, \iota, a)$$

where

$$N \in \mathbb{N}, \qquad \iota : I \times \mathrm{Fin}(2) \to \mathbb{N}, \qquad a : \mathbb{Z} \to R.$$

It is locally nondegenerate when

$$\iota(i, 0) \neq \iota(i, 1) \qquad (i \in I).$$

A committed cochain $c = (q, n, \mathbf{c}, \tau)$ is aligned with $\mathcal{E}$ when

$$c \in \mathcal{C}^{\mathrm{wf}}_{\mathrm{com}}, \qquad n = N, \qquad \iota(i, j) < |\mathbf{c}| \quad (i \in I,\ j \in \mathrm{Fin}(2)).$$

The extracted plane is

$$p_c(i, j) = a(\mathbf{c}_{\iota(i,j)}).$$

A plane target is a pair

$$\mathcal{P} = (\mathcal{C}_{\mathcal{P}}, d_{\mathcal{P}})$$

on plane words. A cochain-to-plane projection satisfies

$$c \in \mathcal{C}_L \Longrightarrow c \text{ is aligned with } \mathcal{E},$$
$$c \in \mathcal{C}_L \Longrightarrow p_c \in \mathcal{C}_{\mathcal{P}},$$
$$d_{\mathcal{P}}(p_c) \le d_L(c).$$

The associated invariants are local nondegeneracy, alignment, well-formedness, expected edge count, selected-index bounds, codeword preservation, and distance control.

Now let F be a finite field. A plane encoding is

$$\Phi : (\mathrm{Fin}(2) \to R) \to F.$$

It sends a plane word p to a received word

$$\Phi_*(p) : I \to F, \qquad \Phi_*(p)(i) = \Phi(p(i, \cdot)).$$

An RS interpretation retains an independently meaningful plane target $(\mathcal{C}_{\mathcal{P}}, d_{\mathcal{P}})$ and supplies the two later-stage conditions

$$p \in \mathcal{C}_{\mathcal{P}} \Longrightarrow \Phi_*(p) \in T, \qquad D(\Phi_*(p)) \le d_{\mathcal{P}}(p).$$

It therefore exposes an evaluation projection from the local syndrome space to the RS code space, with

$$c \in \mathcal{C}_L \Longrightarrow \Phi_*(p_c) \in T, \qquad D(\Phi_*(p_c)) \le d_L(c).$$

For the canonical integer boundary used below,

$$F = \mathbb{Z}/12289\mathbb{Z}, \qquad \Phi(x_0, x_1) = x_0 + g x_1,$$

where g is the canonical RS generator defined in Section 6.7.

This integer encoding is not globally injective. If the first coordinate is shifted by 12289, its image in F is unchanged. Consequently the plane-to-RS far theorem cannot be inferred from the displayed encoding alone. It must either restrict plane points to an admissible region on which Φ is faithful, or supply the required metric lower bound directly. For a set $A \subseteq (\mathrm{Fin}(2) \to \mathbb{Z})$, write

$$\mathrm{FaithfulOn}(\Phi, A) \quad \Longleftrightarrow \quad \Phi|_A \text{ is injective.}$$

Coordinatewise membership in A, together with $\mathrm{FaithfulOn}(\Phi, A)$, prevents two plane words from encoding to the same received word. This is a non-collapse precondition, not yet a quantitative distance lower bound.

Proposition 6.10 (Plane and RS handoff)**.** *Let $c \in \mathcal{C}_{\mathrm{com}}$. If $c \in \mathcal{C}_L$, then the cochain-plane projection gives an aligned plane p_c, and the plane-to-RS interpretation gives*

$$\Phi_*(p_c) \in T, \qquad D(\Phi_*(p_c)) \le d_L(c).$$

Proof. The cochain-to-plane projection has three defining hypotheses. Applied to $c \in \mathcal{C}_L$, they give alignment of c with $\mathcal{E}$, the target-codeword assertion for p_c, and

$$d_{\mathcal{P}}(p_c) \le d_L(c).$$

The plane-to-RS interpretation maps the resulting plane codeword to an RS codeword and gives

$$D(\Phi_*(p_c)) \le d_{\mathcal{P}}(p_c).$$

Composing this inequality with $d_{\mathcal{P}}(p_c) \le d_L(c)$ gives the metric conclusion. □

6.5 Composed Chain, Gates, And Evidence

A concrete projection chain is a pair

$$\Gamma = (\pi_{\mathrm{Pl}\to c}, \pi_{c\to \mathrm{RS}})$$

consisting of the Plucker-to-cochain projection and the plane-to-RS interpretation. For a frame witness $\mathcal{F}$, define

$$\begin{aligned} c_\Gamma(\mathcal{F}) &= E(\mathrm{Pl}(\mathcal{F})), \\ p_\Gamma(\mathcal{F}) &= p_{c_\Gamma(\mathcal{F})}, \\ y_\Gamma(\mathcal{F}) &= \Phi_*(p_\Gamma(\mathcal{F})). \end{aligned}$$

The composed evaluation projection is

$$\mathcal{F} \longmapsto y_\Gamma(\mathcal{F}).$$

Its metric handoff is the chain of inequalities

$$D(y_\Gamma(\mathcal{F})) \le d_{\mathcal{P}}(p_\Gamma(\mathcal{F})) \le d_L(c_\Gamma(\mathcal{F})) \le \rho(\mathcal{F}).$$

Proposition 6.11 (Composed projection chain)**.** *If $\mathcal{F}$ is a valid Plucker frame, then*

$$y_\Gamma(\mathcal{F}) \in T.$$

For every frame witness $\mathcal{F}$,

$$D(y_\Gamma(\mathcal{F})) \le \rho(\mathcal{F}).$$

Proof. For a valid frame, the Plucker-to-cochain projection gives

$$c_\Gamma(\mathcal{F}) \in G_S.$$

The local syndrome target then gives $\sigma(c_\Gamma(\mathcal{F})) \in Z$, so

$$c_\Gamma(\mathcal{F}) \in \mathcal{C}_L.$$

The plane-to-RS interpretation maps local codewords to RS codewords; hence

$$y_\Gamma(\mathcal{F}) = \Phi_*(p_\Gamma(\mathcal{F})) \in T.$$

For the metric statement, the plane-to-RS interpretation gives

$$D(y_\Gamma(\mathcal{F})) \le d_{\mathcal{P}}(p_\Gamma(\mathcal{F})),$$

the cochain-to-plane projection gives

$$d_{\mathcal{P}}(p_\Gamma(\mathcal{F})) \le d_L(c_\Gamma(\mathcal{F})),$$

and the Plucker-to-cochain projection gives

$$d_L(c_\Gamma(\mathcal{F})) \le \rho(\mathcal{F}).$$

The desired inequality follows by transitivity. □

For one witness, the state bus is nested in the same order as the projection: the Plucker/cochain state stores $c_\Gamma(\mathcal{F})$ and $d_L(c_\Gamma(\mathcal{F})) \le \rho(\mathcal{F})$; the cochain/plane state stores that predecessor together with $p_\Gamma(\mathcal{F})$ and $d_{\mathcal{P}}(p_\Gamma(\mathcal{F})) \le d_L(c_\Gamma(\mathcal{F}))$; and the final plane/RS state stores its predecessor together with $y_\Gamma(\mathcal{F})$ and $D(y_\Gamma(\mathcal{F})) \le d_{\mathcal{P}}(p_\Gamma(\mathcal{F}))$. The definitional equalities record that each output is exactly the result of its owning transition. Transitivity yields

$$D(y_\Gamma(\mathcal{F})) \le \rho(\mathcal{F}).$$

The accumulated stage certificate is ordered:

$$\chi_{\mathrm{Pl}\to c} \longrightarrow \chi_{c\to p} \longrightarrow \chi_{p\to \mathrm{RS}} \longrightarrow \chi_{\mathrm{stage}}.$$

The final stage certificate is equivalent to the three explicit invariant conditions for the Plucker/cochain, cochain/plane, and plane/RS stages.

Definition 6.12 (Stage certificates)**.** *For a chain* Γ*, let*

$$\chi_{\mathrm{Pl}\to c}(\Gamma)$$

denote the conjunction

$$\begin{gathered}\mathcal{F} \in \mathcal{V}_{\mathrm{fr}} \Rightarrow c_\Gamma(\mathcal{F}) \in G_S,\\ \mathcal{F} \in \mathcal{V}_{\mathrm{fr}} \Rightarrow c_\Gamma(\mathcal{F}) \in \mathcal{C}_L,\\ d_L(c_\Gamma(\mathcal{F})) \le \rho(\mathcal{F}).\end{gathered}$$

Let

$$\chi_{c\to p}(\Gamma)$$

denote the conjunction of the typed restriction and syndrome naturality squares for the cochain-to-plane transition, local nondegeneracy of $\mathcal{E}$*, alignment of all local codewords, well-formedness of aligned codewords, expected edge-count equality, selected-index bounds, plane-target codeword preservation, and*

$$d_{\mathcal{P}}(p_c) \le d_L(c).$$

Let

$$\chi_{p\to \mathrm{RS}}(\Gamma)$$

denote the conjunction

$$\begin{gathered}\text{the typed restriction and syndrome naturality squares for } p \to \mathrm{RS},\\ \Phi_*(p_c) \text{ is the received word associated with } c,\\ p \in \mathcal{C}_{\mathcal{P}} \Rightarrow \Phi_*(p) \in T,\\ D(\Phi_*(p)) \le d_{\mathcal{P}}(p),\end{gathered}$$

together with the resulting composed codeword and distance consequences for local cochains. The accumulated stage certificate is the conjunction

$$\chi_{\mathrm{stage}}(\Gamma) = \chi_{\mathrm{Pl}\to c}(\Gamma) \wedge \chi_{c\to p}(\Gamma) \wedge \chi_{p\to \mathrm{RS}}(\Gamma).$$

Proposition 6.13 (Stage certificate existence)**.** *Every projection chain* Γ *satisfying the preceding definitions carries* $\chi_{\mathrm{stage}}(\Gamma)$*.*

Proof. The Plucker-to-cochain projection supplies exactly the first and third clauses of $\chi_{\mathrm{Pl}\to c}$. The second follows because an accepted committed cocycle is a local codeword by the defining implication

$$c \in G_S \Rightarrow \sigma(c) \in Z.$$

The cochain-to-plane projection hypotheses are precisely its two typed naturality squares, local nondegeneracy, alignment, plane-target preservation, and the distance inequality; the well-formedness, edge-count, and selected-index clauses are consequences of the alignment condition. The plane-to-RS interpretation hypotheses give its two typed naturality squares, the received-word equality, plane-codeword preservation, and the plane-to-RS distance inequality. Taking these three conjunctions gives $\chi_{\mathrm{stage}}(\Gamma)$. □

The structural hard-gate bundle consists of the following four gates:

source metric:	$0 \le \rho(\mathcal{F})$ and ρ is Plucker-quotient invariant,
Plucker metric:	valid frames lie in the selected projective chart and d_{Pl} is quotient invariant,
quotient respect:	$\mathrm{Pl}(\mathcal{F}) = \mathrm{Pl}(\mathcal{G}) \Rightarrow c_\Gamma(\mathcal{F}) = c_\Gamma(\mathcal{G}),\ p_\Gamma(\mathcal{F}) = p_\Gamma(\mathcal{G}),\ y_\Gamma(\mathcal{F}) = y_\Gamma(\mathcal{G})$,
semantic alignment:	valid frames give aligned cochains, and the plane and RS words are exactly the extraction and encoding outputs.

Commitment acceptance has already established $c \in G_S$ before the projection chain begins. Thus no converse inference from a local syndrome to sheaf acceptance is required here. The optional assertion $\sigma(c) \in Z \Rightarrow c \in G_S$ would be a separate adequacy hypothesis, not a hard gate of this construction.

Definition 6.14 (Hard-gate certificate)**.** *A hard-gate certificate for* Γ *is the tuple*

$$\chi_{\mathrm{hard}}(\Gamma) = \big(\chi_{\mathrm{stage}}(\Gamma),\ \gamma_{\mathrm{src}},\ \gamma_{\mathrm{Pl}},\ \gamma_{\mathrm{quot}},\ \gamma_{\mathrm{align}}\big),$$

where the four γ*-terms are respectively the source metric gate, Plucker metric/chart gate, Plucker-quotient respect gate, and semantic-alignment gate.*

The two local restriction squares compose to a whole-chain restriction square, and the two local syndrome squares compose to a whole-chain coboundary/syndrome square. The naturality certificate

$$\chi_{\mathrm{nat}}(\Gamma) = \big(\chi_{\mathrm{hard}}(\Gamma), \nu_\Gamma\big)$$

stores the hard certificate unchanged and appends only these two composites. It does not run either stage-local check a second time.

Proposition 6.15 (Hard-gate certificate projections)**.** *A hard-gate certificate exposes the accumulated stage certificate and each of the four structural gates. Its quotient-respect entry implies*

$$\mathrm{Pl}(\mathcal{F}) = \mathrm{Pl}(\mathcal{G}) \Longrightarrow y_\Gamma(\mathcal{F}) = y_\Gamma(\mathcal{G}).$$

Extending it through $\chi_{\mathrm{nat}}(\Gamma)$ exposes the two whole-chain naturality equations obtained by composing the stage-local squares.

Proof. The structural statements are projections from the displayed tuple. The quotient-respect entry contains equality of the downstream received words whenever the Plucker coordinates agree, which is the displayed conclusion. For naturality, first apply the cochain-to-plane restriction square and then the plane-to-RS restriction square; functoriality of Φ_* gives the whole-chain restriction equation. Likewise, first apply the plane-to-RS syndrome square and then the cochain-to-plane syndrome square; applying the intervening syndrome map gives the whole-chain coboundary/syndrome equation. These are compositions of the stored local proofs. □

The far-side gates are not injectivity claims. They are calibrated non-collapse implications:

$$r_S \le \rho(\mathcal{F}) \Longrightarrow r_C \le d_L(c_\Gamma(\mathcal{F})),$$

$$r_C \le d_L(c_\Gamma(\mathcal{F})) \Longrightarrow r_P \le d_{\mathcal{P}}(p_\Gamma(\mathcal{F})),$$

and

$$r_P \le d_{\mathcal{P}}(p_\Gamma(\mathcal{F})) \Longrightarrow r_T \le D(y_\Gamma(\mathcal{F})).$$

Together they yield the far-region separation witness

$$r_S \le \rho(\mathcal{F}) \Longrightarrow r_T \le D(y_\Gamma(\mathcal{F})).$$

If two source witnesses collapse to the same final received word, the theorem still says that any collapsed image of a far source witness is target-far. It rules out far-to-near collapse, not all quotient collapse.

Proposition 6.16 (Far-gate composition)**.** *The three far-side gates imply the final far-region separation*

$$r_S \le \rho(\mathcal{F}) \Longrightarrow r_T \le D(y_\Gamma(\mathcal{F})).$$

If $y_\Gamma(\mathcal{F}) = y_\Gamma(\mathcal{G})$ and $r_S \le \rho(\mathcal{F})$, then

$$r_T \le D(y_\Gamma(\mathcal{G})).$$

Proof. Assume $r_S \le \rho(\mathcal{F})$. The first far gate gives

$$r_C \le d_L(c_\Gamma(\mathcal{F})).$$

The second gives

$$r_P \le d_{\mathcal{P}}(p_\Gamma(\mathcal{F})),$$

and the third gives

$$r_T \le D(y_\Gamma(\mathcal{F})).$$

If $y_\Gamma(\mathcal{F}) = y_\Gamma(\mathcal{G})$, substitute this equality in the last inequality to obtain $r_T \le D(y_\Gamma(\mathcal{G}))$. □

A radius calibration is

$$\kappa = (r^S_{\mathrm{near}}, r^S_{\mathrm{far}}, r^T_{\mathrm{near}}, r^T_{\mathrm{far}})$$

with source gap, target gap, and near compatibility. A calibrated final certificate consists of a naturality-extended hard-gate certificate and a far-region separation witness at the calibrated far radii. The intermediate cochain and plane far radii remain explicit in a three-stage far pipeline, but the final certificate records only its calibrated source and target endpoints. It yields calibrated end-to-end readiness and hence a calibrated gap projection into the RS code space.

Definition 6.17 (Calibrated final certificate)**.** *For a radius calibration κ, a calibrated final certificate is*

$$\Omega(\Gamma, \kappa) = \big(\chi_{\mathrm{nat}}(\Gamma), \omega_\kappa\big),$$

where ω_κ is the far-region separation statement

$$r^S_{\mathrm{far}} \le \rho(\mathcal{F}) \Longrightarrow r^T_{\mathrm{far}} \le D(y_\Gamma(\mathcal{F})).$$

The associated readiness condition is the conjunction

$$\chi_{\mathrm{stage}}(\Gamma) \wedge r^S_{\mathrm{near}} \le r^T_{\mathrm{near}} \wedge \omega_\kappa.$$

Proposition 6.18 (Final certificate gives readiness)**.** *A calibrated final certificate $\Omega(\Gamma, \kappa)$ gives the readiness condition above, and therefore gives the calibrated end-to-end gap projection of Theorem 6.2.*

Proof. Project $\Omega(\Gamma, \kappa)$ onto its naturality-extended hard-gate certificate and its far-region witness. By Theorem 6.15, the former retains the hard-gate certificate, which exposes $\chi_{\mathrm{stage}}(\Gamma)$. The near compatibility

$$r^S_{\mathrm{near}} \le r^T_{\mathrm{near}}$$

is part of the radius calibration κ, and the far non-collapse implication is ω_κ. These are exactly the three clauses of the readiness condition. Combining readiness with the evaluation projection of Theorem 6.11 gives the calibrated gap projection, and Theorem 6.2 proves the solver-pullback consequence. □

The evidence database has seven fixed slots:

0	accumulated stage certificate
1	source metric gate
2	Plucker metric/chart gate
3	Plucker-quotient respect gate
4	semantic-alignment gate
5	assembled restriction/syndrome naturality gate
6	calibrated final far-region witness.

Completeness means that every required slot binds to its digest and is included under the shared evidence root. In this paper the evidence root is a finite index for the displayed certificate assertions: each slot names one of the mathematical propositions above, and evidence completeness asserts that all such proofs are present. Canonical byte encoding, domain separation, hash binding, replay/state indexing, and live verification belong to the subsequent protocol paper; they are not hypotheses in the mathematical arguments below.

Proposition 6.19 (Evidence completeness)**.** *If the hard-gate prefix is complete, then every hard-gate slot binds to its digest and is included under the evidence root. If the naturality slot is complete, then the composed naturality proof binds and is included under that root. If the final witness slot is complete, then the calibrated far-region witness also binds and is included under the same root. A complete final evidence certificate therefore recovers the calibrated final certificate and calibrated end-to-end readiness.*

Proof. Hard-gate completeness is the conjunction

$$\Big(\forall s \in \{0, \dots, 4\},\ \beta_s \wedge \iota_s\Big) \wedge \beta_{\mathrm{root}}.$$

Taking the corresponding conjuncts gives binding and inclusion for each hard slot. Naturality completeness is

$$\beta_5 \wedge \iota_5,$$

and final-witness completeness is

$$\beta_6 \wedge \iota_6,$$

so the two appended-slot assertions follow. A final evidence certificate contains the mathematical hard-gate certificate, its derived whole-chain

naturality proof, and the final far-region witness. The definition of the calibrated final certificate adjoins the last witness to the naturality-extended certificate, and the calibrated-readiness map is the definition of end-to-end readiness for that final certificate. □

The public plane view is derived from the observed cochain artifact under the same storage stack root. It is specified by the storage root, the cochain artifact digest, inclusion of the cochain artifact, inclusion of the derived plane view, alignment with $\mathcal{E}$, and the equality

$$p = p_c.$$

Consequently, selected plane coordinates are in bounds and the cochain digest binds the artifact being projected.

Proposition 6.20 (Public plane-view certificate)**.** *A public plane-view certificate exposes*

$$\begin{gathered} \beta_{\mathrm{stack}}, \qquad \iota_c, \qquad \iota_p, \\ p = p_c, \qquad c \in \mathcal{C}^{\mathrm{wf}}_{\mathrm{com}}, \qquad c \text{ is aligned with } \mathcal{E}, \\ n_{\mathrm{edge}}(c) = N, \qquad \iota(i,j) < |\mathbf{c}|, \qquad \beta_c. \end{gathered}$$

Proof. The certificate is the conjunction of the storage-root binding β_{stack}, the inclusion assertions ι_c, ι_p, the observed proof-carrying cochain statement, the plane word, the alignment proof, and the equality $p = p_c$. Taking conjuncts gives root binding, both inclusion facts, alignment, and $p = p_c$. The observed cochain statement contains a well-formed proof-carrying commitment, hence $c \in \mathcal{C}^{\mathrm{wf}}_{\mathrm{com}}$. Alignment with $\mathcal{E}$ contains

$$n_{\mathrm{edge}}(c) = N \quad \text{and} \quad \iota(i,j) < |\mathbf{c}|.$$

Finally, well-formedness of the observed proof-carrying artifact includes the digest-binding assertion β_c. □

6.6 Finite Codes, RS Targets, And MCA

For a finite coordinate type I and alphabet A, a word is a function

$$w : I \to A,$$

and a block code is a subset $\mathcal{C} \subseteq A^I$. The Hamming distance is

$$d_H(u,v) = |\{i \in I : u_i \neq v_i\}|,$$

and the relative distance is

$$\delta_H(u,v) = \frac{d_H(u,v)}{|I|}.$$

The Hamming and relative balls around a received word w are

$$B_{\mathcal{C}}(w,r) = \{c \in \mathcal{C} : d_H(c,w) \le r\},$$
$$B_{\mathcal{C}}(w,\delta) = \{c \in \mathcal{C} : \delta_H(c,w) \le \delta\}.$$

The list-size inequalities are

$$\forall w,\ |B_{\mathcal{C}}(w,r)| \le L, \qquad \forall w,\ |B_{\mathcal{C}}(w,\delta)| \le L.$$

A count-radius closeness witness implies relative-radius closeness at $r/|I|$, and the relative-radius map is monotone.

For a field F, an F-additive code is closed under zero, addition, and scalar multiplication. Interleaving is defined by

$$\mathcal{C}^{\oplus T} = \{w : I \to (T \to A) : \forall t \in T,\ i \mapsto w(i,t) \in \mathcal{C}\}.$$

The pair word of $f_1, f_2 \in A^I$ is

$$(f_1,f_2)(i,0) = f_1(i), \qquad (f_1,f_2)(i,1) = f_2(i).$$

An affine line of received words is

$$\ell(\gamma) = f_1 + \gamma f_2.$$

For $S \subseteq I$, write

$$S \text{ is } \delta\text{-large} \iff |S|/|I| \ge 1-\delta.$$

The paper Definition 4.3 MCA bad challenges are

$$\mathrm{Bad}_{\mathrm{MCA}}(\mathcal{C},\delta,\ell) = \{\gamma \in F : \exists S \subseteq I,\ S \text{ is } \delta\text{-large} \wedge \ell(\gamma) \text{ agrees on } S \text{ with some } c \in \mathcal{C} \wedge (f_1,f_2) \text{ does not agree on } S \text{ with } \mathcal{C}^{\oplus 2}\}.$$

The paper-facing MCA error bound is

$$\forall \ell, \qquad \frac{|\,\mathrm{Bad}_{\mathrm{MCA}}(\mathcal{C},\delta,\ell)|}{|F|} \le \varepsilon.$$

The count-based MCA condition implies the count-based correlated-agreement condition: if too many line points are individually close, one of them is MCA-good, and its agreement witness becomes a common agreement witness for the line.

Proposition 6.21 (MCA implies correlated agreement)**.** *Let $\mathcal{C}$ be a finite F-additive block code. If $\mathcal{C}$ satisfies the count-based MCA bound at point radius r_p, line radius r_ℓ, and threshold τ, then $\mathcal{C}$ satisfies the count-based correlated-agreement bound at the same radii and threshold.*

Proof. Fix an affine line ℓ. Let

$$B = \{\gamma \in F : \ell(\gamma) \text{ is close on some large set at radius } r_p\}$$

and let

$$M = \{\gamma \in F : \gamma \text{ is not MCA-good for } \ell\}.$$

The MCA bound says $|M| \le \tau$. Suppose $\tau < |B|$ but correlated agreement fails, so no single large set witnesses line-close agreement at radius r_ℓ. If $\gamma \in B$, choose a large set S witnessing the individual closeness of $\ell(\gamma)$. If γ were MCA-good, this same S would extend to a common line witness, contrary to the assumed failure of correlated agreement. Hence $\gamma \in M$. Thus $B \subseteq M$, so

$$|B| \le |M| \le \tau,$$

contradicting $\tau < |B|$. Therefore a common large agreement set exists, which is precisely the correlated-agreement conclusion. □

For Reed–Solomon, an evaluation domain is an injective map

$$a : I \hookrightarrow F.$$

Parameters (a, k) define the scalar RS code

$$\mathrm{RS}(a, k) = \{(p(a_i))_{i \in I} : \deg p < k\}.$$

The paper alphabet $C \subseteq (F^s)^n$ is the s-fold interleaving of the scalar code. RS target data consist of the parameters, F-additivity, a smooth-regime condition, list radius and list size, point and line radii, MCA threshold, list-decoding bounds, MCA bounds, the paper Definition 4.3 MCA bound, a received-word metric, radius normalization, and bridge conditions. The bridge conditions are

domain cardinality, smooth domain, evaluation-point compatibi
restriction naturality, coboundary/syndrome naturality, global sections map t
projection equals the received word, metric normalization, near-distance compatibility, far non-c

These conditions give the calibrated hcrypt-to-RS bridge, and a correct RS gap solver pulls back to a correct source-side gap solver.

6.7 Canonical Reed–Solomon Endpoint

The canonical endpoint fixes

$$F = \mathbb{Z}/12289\mathbb{Z}, \qquad n = 4096, \qquad k = 1024.$$

Thus the rate is $1/4$. The multiplicative group order is

$$12289 - 1 = 12288 = 3 \cdot 4096.$$

The element $11 \in F^\times$ is fixed as a generator of order 12288. Cubing it gives

$$g = 11^3,$$

with

$$\mathrm{ord}(g) = 4096.$$

The evaluation domain is

$$a_i = g^i, \qquad i \in \mathrm{Fin}(4096),$$

and the order computation proves that $i \mapsto a_i$ is injective. The smoothness witness is the divisibility

$$4096 \mid 12289 - 1.$$

Proposition 6.22 (Canonical power domain)**.** *In $F = \mathbb{Z}/12289\mathbb{Z}$, the element* 11 *has multiplicative order* 12288*, the element*

$$g = 11^3 = 1331$$

has multiplicative order 4096*, and the map*

$$i \longmapsto g^i, \qquad i \in \mathrm{Fin}(4096),$$

is injective.

Proof. The field has $|F^\times| = 12288$. Also

$$12288 = 2^{12} \cdot 3,$$

so the only prime divisors of 12288 are 2 and 3. Direct modular arithmetic gives

$$11^{12288} \equiv 1, \qquad 11^{6144} \equiv 12288 = -1, \qquad 11^{4096} \equiv 6240 \pmod{12289}.$$

Thus $11^{12288} = 1$, while $11^{12288/2} \neq 1$ and $11^{12288/3} \neq 1$. The standard order criterion for a finite group element therefore gives

$$\operatorname{ord}(11) = 12288.$$

The order formula for powers gives

$$\operatorname{ord}(11^3) = \frac{\operatorname{ord}(11)}{\gcd(\operatorname{ord}(11), 3)} = \frac{12288}{3} = 4096.$$

Now suppose $g^i = g^j$ with $0 \leq i, j < 4096$. Then

$$g^{i-j} = 1,$$

so $i \equiv j \pmod{\operatorname{ord}(g)}$, hence

$$i \equiv j \pmod{4096}.$$

The representatives i, j already lie in $\{0, \ldots, 4095\}$, so $i = j$. □

The conservative Johnson-below radius is

$$r_J = 2047, \qquad 2r_J < 4096,$$

and the strict capacity-below radius recorded separately is

$$r_{\text{cap}} = 3071, \qquad 4r_{\text{cap}} < 3 \cdot 4096.$$

The canonical real radius is

$$\delta_J = \frac{2047}{4096}.$$

The canonical metric is relative Hamming distance to the RS code:

$$D(w) = \frac{1}{4096} \inf\{d_H(c, w) : c \in \mathrm{RS}(a, 1024)\}.$$

Exact RS codewords have $D(w) = 0$, and Hamming-close words satisfy the corresponding relative-radius bound. This supplies the radius-normalization object for the canonical target.

Proposition 6.23 (Canonical metric normalization)**.** *For the canonical RS target:*

$$w \in \mathrm{RS}(a, 1024) \Longrightarrow D(w) = 0.$$

If w is Hamming-close to the code at radius r, then

$$D(w) \leq \frac{r}{4096}.$$

Moreover the radius conversion $r \mapsto r/4096$ sends 0 to 0 and is monotone.

Proof. The zero polynomial has degree < 1024, so the zero word is an RS codeword. Thus the set

$$\{d_H(c, w) : c \in \mathrm{RS}(a, 1024)\}$$

is nonempty and bounded below by 0. If w itself is a codeword, this set contains $d_H(w, w) = 0$; since all Hamming distances are nonnegative, its infimum is 0, and hence $D(w) = 0/4096 = 0$.

If w is Hamming-close at radius r, then there is $c \in \mathrm{RS}(a, 1024)$ with $d_H(c, w) \le r$. The infimum of the distance set is at most this displayed distance, so after dividing by 4096 we obtain

$$D(w) \le r/4096.$$

The map $r \mapsto r/4096$ sends 0 to 0, and monotonicity follows from division by the positive integer 4096. □

The canonical bridge data are

$$\beta = (\text{canonical smooth domain},\ \text{canonical RS target},\ D,\ \Phi,\ \delta_J),$$

where $\Phi(x_0, x_1) = x_0 + gx_1$. The endpoint proves the concrete domain, smoothness, evaluation-point, metric-normalization, and radius-matching facts. The remaining semantic theorem slots are:

restriction naturality, coboundary/syndrome naturality, global sections map to RS codewords, projection equals the received word, near-distance compatibility, far non-collapse.

Definition 6.24 (Canonical bridge certificate)**.** *For a canonical chain Γ, let Θ_Γ be the six-tuple of theorem slots*

$$\theta_\Gamma^{\mathrm{rest}},\quad \theta_\Gamma^{\mathrm{cob}},\quad \theta_\Gamma^{\mathrm{code}},$$
$$\theta_\Gamma^{\mathrm{word}},\quad \theta_\Gamma^{\mathrm{near}},\quad \theta_\Gamma^{\mathrm{far}}.$$

A bridge certificate is a proof of their conjunction:

$$\mathcal{B}_\Gamma = \theta_\Gamma^{\mathrm{rest}} \wedge \theta_\Gamma^{\mathrm{cob}} \wedge \theta_\Gamma^{\mathrm{code}} \wedge \theta_\Gamma^{\mathrm{word}} \wedge \theta_\Gamma^{\mathrm{near}} \wedge \theta_\Gamma^{\mathrm{far}}.$$

An RS metric certificate is a pair

$$\Omega_{\mathrm{RS}} = (\kappa, \Omega(\Gamma, \kappa)),$$

where $\Omega(\Gamma, \kappa)$ is the calibrated final certificate of Theorem 6.17.

Proposition 6.25 (RS metric certificate fills the bridge slots)**.** *An RS metric certificate Ω_{RS} determines a bridge certificate for Θ_Γ. Consequently it completes the generic canonical RS bridge conditions once the endpoint data*

$$(F, n, k, a, D, \Phi, \delta_J)$$

have been fixed as above.

Proof. Let $\Omega_{\mathrm{RS}} = (\kappa, \Omega)$. By Theorem 6.18, Ω exposes the accumulated stage certificate, the derived whole-chain naturality gate, near compatibility, and far non-collapse. The naturality gate supplies the restriction-naturality and coboundary/syndrome-naturality slots by composing the two corresponding stage-local squares; it does not assume new unrelated propositions. By Theorem 6.11, valid global frames map to RS codewords, giving $\theta_\Gamma^{\mathrm{code}}$. The projection-matches-word slot is the definitional equality

$$\Gamma(\mathcal{F}) = y_\Gamma(\mathcal{F}).$$

The near slot is κ's near compatibility, and the far slot is Ω's far-region witness. These six proofs form $\mathcal{B}_\Gamma$.

The remaining canonical bridge conditions are endpoint facts: domain cardinality, smoothness, evaluation-point compatibility, metric normalization, and radius matching. They were fixed in the canonical endpoint data and proved in Theorems 6.22 and 6.23. Combining those endpoint facts with $\mathcal{B}_\Gamma$ completes the bridge conditions. □

A canonical Johnson far-separation witness is the statement

$$\delta_J \le \rho(\mathcal{F}) \Longrightarrow \delta_J \le D(y_\Gamma(\mathcal{F})).$$

It may be supplied directly, by a pointwise lower bound

$$\rho(\mathcal{F}) \le D(y_\Gamma(\mathcal{F})),$$

or by exact distance preservation

$$D(y_\Gamma(\mathcal{F})) = \rho(\mathcal{F}).$$

Together with the naturality-extended hard-gate certificate, this witness gives the canonical calibrated RS bridge.

Proposition 6.26 (Canonical bridge completion)**.** *A naturality-extended hard-gate certificate for Γ, together with a canonical Johnson far-separation witness*

$$\delta_J \le \rho(\mathcal{F}) \Longrightarrow \delta_J \le D(y_\Gamma(\mathcal{F})),$$

gives a completed calibrated hcrypt-to-RS bridge for the canonical endpoint.

Proof. The canonical endpoint supplies the concrete bridge conditions: domain cardinality is $|\mathrm{Fin}(4096)| = 4096$, smoothness is $4096 \mid 12289 - 1$, evaluation points are $a_i = g^i$ by Theorem 6.22, the metric normalization is Theorem 6.23, and the radius parameters are all the relative Hamming radius δ_J.

The naturality-extended certificate contains the hard-gate certificate, hence the accumulated stage certificate and structural gate bundle. Its two naturality fields are the whole-chain composites of the restriction and syndrome squares already stored at the cochain/plane and plane/RS stages. By Theorem 6.11, valid source frames map to RS codewords and the composed map is definitionally $\mathcal{F} \mapsto y_\Gamma(\mathcal{F})$. The same-radius calibration has near compatibility by reflexivity:

$$\delta_J \le \delta_J.$$

The displayed far-separation witness supplies the final far non-collapse slot. The naturality-extended certificate and far-separation witness form the calibrated final certificate $\Omega(\Gamma, \kappa)$; together with κ, this is the RS metric certificate Ω_{RS}. By Theorem 6.25, it fills the bridge theorem slots and completes the canonical RS bridge conditions. Then Theorem 6.2 gives the induced pullback of correct RS gap solvers. □

The deep list-decoding and MCA estimates are external theorem inputs [BSCH+26, ABF26].

Assumption 6.27 (Cited canonical list/MCA bound)**.** *The cited theorem boundary supplies:*

$$\begin{gathered} L_{\mathrm{cit}} \in \mathbb{N}, \qquad \tau_{\mathrm{cit}} \in \mathbb{N}, \\ \mathcal{C}_{\mathrm{RS}} \text{ is } (r_J, L_{\mathrm{cit}})\text{-list decodable}, \\ \mathcal{C}_{\mathrm{RS}} \text{ is } (\delta_J, L_{\mathrm{cit}})\text{-list decodable}, \\ \mathcal{C}_{\mathrm{RS}} \text{ satisfies } (r_J, r_J, \tau_{\mathrm{cit}})\text{-MCA}, \\ \varepsilon_{\mathrm{MCA}} \le \frac{\tau_{\mathrm{cit}}}{|F|}, \qquad \varepsilon_{\mathrm{MCA}}^{\mathrm{paper}}(\mathcal{C}_{\mathrm{RS}}, \delta_J) \le \frac{\tau_{\mathrm{cit}}}{|F|}. \end{gathered}$$

Here τ_{cit} and L_{cit} denote the specialized finite witnesses. This paper does not claim $\tau_{\mathrm{cit}} < |F|$, or any fixed-bit security level, without a separate numerical evaluation of the cited formula at the canonical parameters. Thus the displayed fraction is a theorem-shaped external bound; its nontriviality is a distinct parameter check.

6.8 Canonical MCA Corollary

For the canonical scalar RS endpoint, let

$$\mathcal{W}_{\mathrm{fin}} = F^{\mathrm{Fin}(4096)}.$$

A canonical MCA line is

$$L = (f_1, f_2), \qquad L(\gamma) = f_1 + \gamma f_2.$$

The projection chain induces such a line from two source frame witnesses:

$$L_{\Gamma,\mathcal{F}_1,\mathcal{F}_2} = (y_\Gamma(\mathcal{F}_1), y_\Gamma(\mathcal{F}_2)).$$

The canonical bad-challenge set is

$$\mathrm{Bad}_\Gamma(\mathcal{F}_1, \mathcal{F}_2) = \mathrm{Bad}_{\mathrm{MCA}}(\mathrm{RS}(a, 1024), \delta_J, L_{\Gamma,\mathcal{F}_1,\mathcal{F}_2}).$$

The paper-alphabet endpoint $\mathcal{C}_{\mathrm{RS}} \subseteq (F^s)^n$ specializes here to $s = 1$. At width one,

$$F \cong F^1, \qquad F^{4096} \cong (F^1)^{4096},$$

and the scalar canonical bad-challenge set agrees with the paper's $s = 1$ bad-challenge set. Therefore the imported paper Definition 4.3 MCA bound yields the scalar canonical bound.

Theorem 6.28 (Canonical RS MCA bound)**.** *For the canonical Homotopical Encryption Reed–Solomon endpoint at radius $\delta_J = 2047/4096$, every projection-induced final affine line satisfies*

$$\frac{|\,\mathrm{Bad}_\Gamma(\mathcal{F}_1, \mathcal{F}_2)|}{|F|} \leq \frac{\tau_{\mathrm{cit}}}{|F|}.$$

Proof. Let

$$L = L_{\Gamma,\mathcal{F}_1,\mathcal{F}_2} = (y_\Gamma(\mathcal{F}_1), y_\Gamma(\mathcal{F}_2)).$$

By definition,

$$\mathrm{Bad}_\Gamma(\mathcal{F}_1, \mathcal{F}_2) = \mathrm{Bad}_{\mathrm{MCA}}(\mathrm{RS}(a, 1024), \delta_J, L).$$

It remains only to compare the scalar canonical line L with the paper alphabet at width $s = 1$. The maps

$$F \longrightarrow F^1, \qquad a \longmapsto (* \mapsto a),$$

and

$$F^1 \longrightarrow F, \qquad u \longmapsto u(*),$$

are inverse. Applying them coordinatewise gives inverse bijections between

$$F^{4096} \quad \text{and} \quad (F^1)^{4096}.$$

Under this identification, scalar RS codewords are exactly width-one paper-alphabet RS codewords: a width-one vector word is a codeword precisely when its unique scalar component is an RS codeword. Agreement on a coordinate set S is also preserved and reflected by the same bijection. The pair word (f_1, f_2) is preserved componentwise, so closeness of the interleaved pair to $C^{\oplus 2}$ on S is unchanged.

Therefore the scalar bad-challenge set is equal to the paper Definition 4.3 bad-challenge set for the width-one endpoint. The cited width-one paper-MCA theorem gives

$$\frac{|\operatorname{Bad}_{\mathrm{MCA}}(\mathrm{RS}(a, 1024), \delta_J, L)|}{|F|} \leq \frac{\tau_{\mathrm{cit}}}{|F|}.$$

Substituting the displayed equality of bad-challenge sets gives the theorem. □

Corollary 6.29 (Verifier-facing RS/MCA projection)**.** *Given an accepted hidden/geometric witness, a certified Plucker-to-cochain-to-plane-to-RS projection chain, hard projection-gate evidence extended through assembled naturality, and a canonical Johnson-radius far-separation witness, the public verifier obtains a finite Reed–Solomon received word whose canonical MCA bad-challenge fraction is bounded by the cited threshold* $\tau_{\mathrm{cit}}/|F|$.

Proof. The certified projection chain provides the maps

$$\mathcal{F} \longmapsto c_\Gamma(\mathcal{F}) \longmapsto p_\Gamma(\mathcal{F}) \longmapsto y_\Gamma(\mathcal{F}).$$

By Theorem 6.11, valid source data map to RS received words and satisfy

$$D(y_\Gamma(\mathcal{F})) \leq \rho(\mathcal{F}).$$

The projection-gate evidence gives the structural and stage certificates and the composed naturality equations; the Johnson far-separation witness then gives the calibrated bridge by Theorem 6.26. For any second projected witness $\mathcal{G}$, Theorem 6.28 applied to the final affine line

$$(y_\Gamma(\mathcal{F}), y_\Gamma(\mathcal{G}))$$

gives the stated bound

$$|\operatorname{Bad}_\Gamma(\mathcal{F},\mathcal{G})|/|F| \le \tau_{\mathrm{cit}}/|F|.$$

□

This corollary is a proximity statement. It says that accepted hidden/geometric state can be projected to a finite public RS/MCA target with calibrated non-collapse at the selected far region. It does not say that the projection is injective, and it does not turn the RS/MCA endpoint into the source of HCP hardness.

Lean Cross-Reference

The end-to-end projection and gap vocabulary corresponds to `Proximity.Projection` and `Proximity.Problem`. The Grassmannian and Plucker source stages correspond to `Proximity.Grassmannian`, `Proximity.GrassmannianPlucker`, and `Proximity.GrassmannianPlucker.Concrete`. The committed-cochain and local-syndrome stages correspond to `Proximity.CochainBridge`, `Proximity.PluckerCochain`, `Proximity.PluckerCochain.Concrete`, and `Proximity.PluckerCochain.PublicBinding`; their accepted-cocycle predicate is `Proximity.CochainBridge.CommittedCochain.AcceptedCocycle`, which inherits `Axioms.CommittedCochainWellFormed` and `Axioms.oneCocycle_of_committedCochainWellFormed`. The plane and finite-field encoding stages correspond to `Proximity.CochainPlane.Core`, `Proximity.CochainPlane.Concrete`, `Proximity.CochainPlane.ReedSolomon`, and `Proximity.CochainReedSolomon`. In particular, `CochainReedSolomon.PlaneEncoding.receivedWord_injective_on` is the faithful-admissible-region lemma, while `CanonicalRS.integerPlaneEncoding_not_injective` records the global modulus collision at the canonical integer boundary.

The composed chain, gates, public boundary, and evidence records correspond to `Proximity.ProjectionChain.Core`, `Proximity.Gates.Stage`, `Proximity.Gates.Final`, `Proximity.Gates`, `Proximity.GateEvidence`, and `Proximity.Public`. The generic finite code, RS, list-decoding, and CA/MCA vocabulary corresponds to `Proximity.RS.ListDecoding`, `Proximity.RS.CorrelatedAgreement`, `Proximity.RS.ReedSolomon`, `Proximity.RS.Target`, and `Proximity.RS`. The canonical endpoint and MCA specialization correspond to `Proximity.CanonicalRS.Parameters`, `Proximity.CanonicalRS.PowerDomain`,

`Proximity.CanonicalRS.Encoding`, `Proximity.CanonicalRS.Target`, `Proximity.CanonicalRS.Metric`, `Proximity.CanonicalRS.BridgeData`, `Proximity.CanonicalRS.BridgeWitness`, `Proximity.CanonicalRS.BridgeCertificate`, `Proximity.CanonicalRS.FarSeparation`, `Proximity.CanonicalRS.MCA.Line`, `Proximity.CanonicalRS.MCA`, `Proximity.CanonicalRS.MCA.Concrete`, `Proximity.CanonicalRS.CitedBounds`, and `Proximity.CanonicalRS`.

7 Future Directions

7.1 Protocol Boundary

The present paper stops at the mathematical and formalization boundary. A subsequent protocol paper should turn the commitment, oracle, proof-carrying, identity, storage, and proximity surfaces into explicit adversary games and executable verification paths. The concrete protocol must specify which certificates are public evidence, which witnesses remain private, and which runtime failures cause rejection, hold, or re-evaluation.

On the identity side, the hidden-representative/public-classifier split should be developed into explicit class-membership, unlinkability, and nullifier properties. Public identity data should remain class values, hashes, proofs, or oracle judgments; the private side should retain fibration representatives, transport data, and the Hodge operator selecting the operative kernel.

On the storage side, the commitment layer can be developed as a controlled system for sheaf/cochain projections. A public store may retain topology hashes, cochain roots, compressed public data, and proof-carrying statements, while full sheaf and fibration witnesses remain private or are opened only through dedicated proof paths.

The protocol-facing reference point is the higher categorical cryptoeconomics program [SKdS25]. It is relevant to the design of protocol evidence and state semantics, but it is not used in the proofs of the present paper.

7.2 Complexity-Theoretic Refinements

The current formalization isolates the hidden-operator obstruction: public data does not canonically determine the operative Hodge–Laplacian kernel. It also separates the external known-lattice CVP assumption from the project-native hidden-operator ambiguity, gives public-data-only oracle insufficiency statements, records classical and quantum adversary likelihood surfaces, and proves infinitude results for the relevant Grassmannian/Laplacian families.

Future work should sharpen these surfaces quantitatively: rates of growth for hidden-kernel families, refined leakage bounds for public classifiers, success probabilities for specified adversary classes, and reductions between variants of HCP.

A Paper–Lean Cross-Reference

A.1 Definitions And Interfaces

HCP public observation and hidden witness.
Sections 2–3; `Hardness . PublicProjection`, `Hardness . HiddenOperatorFiber`, and `Hardness.HiddenOperatorClass`.

Matrix harmonicity and Hodge instances.
Section 2; `Hardness.Hodge_Instance` and `Hardness.MatrixIsHarmonic`.

Kernel ambiguity and lossy public data.
Sections 2–3; `Hardness . SameKernel`, `Hardness . KernelAmbiguous`, `Hardness . PublicCompatibleHodgeAmbiguity`, and `Hardness . LossyPublicProjection`.

Public-data-only Hodge solvers.
Section 3; `Hardness . PublicHodgeInstance`, `Hardness . PublicHodgeSolver`, `Hardness . PublicHodgeSolverSound`, and `Hardness.PublicHodgeSolverExactOnHarmonic`.

Explicit-lattice CVP comparison.
Section 3; `CVP.Instance`, `CVP.Solution`, and `CVP.OracleCorrect`.

Harmonic-lattice bridge.
Section 3; `Hardness . HarmonicCVPBridge` and `Hardness . HarmonicCVPBridge.projectedTarget`.

Closest-harmonic solution predicates.
Section 3; `Hardness . ClosestHarmonicSolution` and `Hardness . ProjectionClosestHarmonicSolution`.

Hodge projection model.
Sections 2–3; `Laplacian.harmonicProjection`, `Laplacian.residual`, and `Laplacian.IsHarmonic`.

Categorical fibration copy surface.
Section 4; `Hardness . IsCartesianMorphism`, `Hardness . GrothendieckFibration`, `Hardness . HomotopyEquivalent`, and `Hardness.FibrationCopy`.

Semantic/public equality surface.
Section 4; `Axioms.HelferFOL`, `Axioms.PublicClassifierFactorization`, `Axioms . PublicClassifierFactorization . publicFiber _ eq _ iUnion _ baseFibers`, `Axioms.HomotopyClass`, and `Axioms.Fibration`.

Heyting/oracle public judgment.
Section 4; `Axioms.Heyting3`, `Axioms.OracleQuery`, `Axioms.OracleAnswer`, and `Axioms.OracleEta2`.

Cochain commitment payload.
Section 5; `Axioms.CochainCommitmentData`, `CochainCommitmentData.WellFormed`, and `CochainCommitmentData.AddCompatible`.

Commitment security interface.
Section 5; `Axioms.CochainCommitmentAxioms`.

Proof-carrying public artifact.
Section 5; `TCB.HashedCommittedCochain` and `TCB.ProofCarryingCommittedCochain`.

Projection chain.
Section 6; `ProjectionChain.Chain` and `ProjectionChain.Chain.toEvaluationProjection`.

Proximity-stage certificates.
Section 6; `Proximity.Gates.SourceMetricGate`, `Proximity.Gates.HardGateCertificate`, `Proximity.Gates.ProjectionNaturalityGate`, and `Proximity.Gates.NaturalityGateCertificate`. The separately defined `Proximity.Gates.SyndromeAdequacyGate` is optional and is not a hard gate of the accepted-commitment projection chain.

Canonical RS/MCA endpoint.
Section 6; `CanonicalRS.BridgeData`, `CanonicalRS.MCA.Line`, and `CanonicalRS.MCA.badChallengeFraction`.

A.2 Theorems And Status

Kernel ambiguity implies lossy public projection.
`Hardness.kernelAmbiguous_implies_lossy`; Lean-proved.

Public-compatible Hodge ambiguity gives kernel ambiguity.
`Hardness.kernelAmbiguous_of_publicCompatibleHodgeAmbiguity`; Lean-proved.

Concrete Hodge ambiguity exists in nontrivial dimension.
`Hardness.Hodge_Ambiguity`, `Hardness.constantPublicProjection_publicCompatibleHodgeAmbiguity`, `Hardness.constantPublicProjection_kernelAmbiguous`, and `Hardness.constantPublicProjection_lossy`; Lean-proved.

Every field-linear image can be a Hodge kernel.
`Hardness.exists_matrix_ker_eq_range`; Lean-proved linear-algebra construction.

Kernel realization for presented generated subspaces.
`Hardness.Hodge_GeneratedSubspace_Reduction` and `Hardness.Hodge_GeneratedSubspace_Reduction_Proof`; Lean-proved surface.

Hodge decomposition into harmonic, exact, and coexact parts.
`Hardness.Hodge_Decomposition_Theorem`; Lean-proved theorem surface.

Ambiguous input contradicts public-data-only Hodge solving.
`Hardness.public_solver_eq_input_of_harmonic` and `Hardness.no_public_solver_on_ambiguous_input`; Lean-proved lemmas.

Known-lattice CVP does not select the hidden operator.
`Hardness.CVP_Oracle_Insufficient_For_Public_Hodge_Solving`; Lean-proved surface.

No uniform public-data Hodge solver.
`Hardness.no_uniform_public_hodge_solver`; Lean-proved hidden-operator model theorem.

Projection-model closest-harmonic surface.
`Hardness.matrix_harmonic_iff_projection_isHarmonic`, `Hardness.Hodge_Projection_Is_ClosestHarmonic`, and `Hardness.harmonicProjection_is_closestHarmonic`; Lean-proved.

Discrete CVP minimizers are preserved by harmonic projection.
`Hardness.HarmonicCVPBridge.squaredDistance_projection_decomposition`, `Hardness.HarmonicCVPBridge.isClosestLatticeVector_iff_projectedTarget`, and `Hardness.HarmonicCVPBridge.solution_iff_projectedTarget`; Lean-proved.

Fibration copies preserve the hidden semantic surface.
`Hardness.copy_equivalent_thm` and `Hardness.FibrationCopy_is_Grothendieck`; Lean-proved categorical interface.

Exact sheaf gates supply $d^{k+1}d^k = 0$ and Laplacian-from-coboundary.
`Gates.structural_preconditions_of_sheaf_exact_gates`; Lean-proved alias/interface.

Well-formed payloads have nonempty edge data and supported modulus.
`CochainCommitmentData.edge_values_nonempty_of_wellFormed` and `CochainCommitmentData.modulus_gt_one_of_wellFormed`; Lean-proved.

Proof-carrying artifacts expose the committed payload.
`TCB.proofCarrying_cochain_wellFormed`; TCB boundary theorem.

Projection chain composes to one evaluation projection.
`ProjectionChain.Chain.toEvaluationProjection`; Lean-proved plumbing.

Syndrome adequacy turns zero local syndrome into codeword/global candidate.
`Proximity.Gates.SyndromeAdequacyGate.codeword_of_wellFormed_zeroSyndrome`; stage-local interface.

Stage naturality composes without rechecking.
`Proximity.Gates.ProjectionNaturalityGate.of_stageCertificate`; Lean-proved composition of the cochain/plane and plane/RS restriction and syndrome squares.

Final proximity evidence yields calibrated end-to-end readiness.
`Proximity.Gates.CalibratedFinalCertificate.toCalibratedEndToEndReady`; Lean-proved plumbing.

Canonical RS list/MCA bounds enter at a cited boundary.
`CanonicalRS.CitedBounds.bounds`; cited external theorem package.

Canonical MCA bad-challenge fraction is bounded after projection.
`CanonicalRS.MCA.projectionBadChallengeFraction_le`; Lean-proved from cited bounds.

B Lean Source Coverage By File

This appendix is the file-by-file coverage audit for the translation. The body of the paper gives the mathematical definitions and proofs; this appendix records where each Lean declaration family enters the written account. Import-only aggregate files are listed with the files they collect.

B.1 Build And Aggregate Files

`lakefile.lean`.
Build configuration; no mathematical declaration is translated as a theorem.

`Axioms.lean`, `Gates.lean`, `TCB.lean`, `Consensus.lean`, `Proximity.lean`, and `Proximity/CanonicalRS.lean`.
Aggregate import surfaces. Their mathematical content is exactly the union of the component files listed below.

B.2 HCP, CVP, And Fibrational Algebra

`Hardness.lean`.
Sections 2–4 translate every declaration into the hidden Hodge-operator algebra. The HCP data are `Hodge_Instance`, `MatrixIsHarmonic`, `PublicProjection`, `PubliclyEquivalent`, `HiddenOperatorFiber`, and `HiddenOperatorClass`. The ambiguity predicates are `LossyPublicProjection`, `SameKernel`, `KernelAmbiguous`, and `PublicCompatibleHodgeAmbiguity`. The elementary closure lemmas `mem_hiddenOperatorClass_self`, `sameKernel_refl`, `kernelAmbiguous_implies_lossy`, and `kernelAmbiguous_of_publicCompatibleHodgeAmbiguity` are Theorem 2.10. The concrete ambiguity theorems `Hodge_Ambiguity`, `Hodge_Ambiguity_Proof`, `constantPublicProjection_`

publicCompatibleHodgeAmbiguity, constantPublicProjection_kernelAmbiguous, and constantPublicProjection_lossy are Theorem 3.4. The presented-linear-image comparison declarations Hodge_Solver_Correct, exists_matrix_ker_eq_range, Hodge_GeneratedSubspace_Reduction, and Hodge_GeneratedSubspace_Reduction_Proof are Theorem 3.1. The decomposition theorem Hodge_Decomposition_Theorem is Theorem 2.5. The public-solver predicates PublicHodgeInstance, PublicHodgeSolver, PublicHodgeSolverSound, PublicHodgeSolverExactOnHarmonic, public_solver_eq_input_of_harmonic, no_public_solver_on_ambiguous_input, no_uniform_public_hodge_solver, and CVP_Oracle_Insufficient_For_Public_Hodge_Solving are Theorem 3.4. The closest-harmonic predicates ClosestHarmonicSolution, ProjectionClosestHarmonicSolution, matrix_harmonic_iff_projection_isHarmonic, Hodge_Projection_Is_ClosestHarmonic, and harmonicProjection_is_closestHarmonic record the continuous projection surface. The discrete bridge declarations HarmonicCVPBridge, HarmonicCVPBridge.squaredDistance_projection_decomposition, HarmonicCVPBridge.isClosestLatticeVector_iff_projectedTarget, and HarmonicCVPBridge.solution_iff_projectedTarget are Theorem 3.2. The categorical declarations IsCartesianMorphism, GrothendieckFibration, HomotopyEquivalent, Wrapper, Fibration, FibrationCopy, ToWrapper, FromWrapper, copy_equivalent_thm, and FibrationCopy_is_Grothendieck are Theorem 4.1.

CVP.lean.
Section 3 translates AmbientVector, ExplicitLattice, latticeVector, InExplicitLattice, IsClosestLatticeVector, squaredEuclideanDistance, Instance, Solution, Oracle, LatticeOracleCorrect, OracleCorrect, SolverCorrect, solverCorrect_iff_oracleCorrect, HardnessAssumption, and zero_lattice_oracle_correct as the known-lattice comparison surface.

Axioms / HelferFOL . lean, Axioms / Fibration . lean, and Axioms/HomotopyClass.lean.
Section 4 translates path equality, public classifiers, public fibers, classifier factorization through a base projection, class-membership witnesses, transport, the Helfer semantic well-formedness clauses, fiber systems, ephemeral fiber specifications, and nonfaithful classifiers. The equivalence, symmetry, transitivity, membership, and nonfaithfulness lemmas justify the public fiber language used in HCP.

B.3 Foundational, Sheaf, Spectral, And Judgment Axioms

Axioms/Foundation.lean and Gates/Exact.lean.
Section 4 translates Foundation and FoundationWellFormed: finite cochain data, coboundary compatibility, Hodge Laplacian formation, topol-

ogy connectivity, spectral gap positivity, and public key-image/freshness premises. The exact-gate aliases `ExactGateBundle`, `SheafExactGateBundle`, `FoundationWithSheafExactGates`, and `structural_preconditions_of_sheaf_exact_gates` are Theorem 4.8.

`Axioms/SheafCoherence.lean`.
Section 4 translates restriction-shape compatibility, edge naturality, triangle-face commutativity, sheaf cochain complexes, Hodge coherence, finite simplicial nerves, `SheafCoherence`, `SheafCoherenceWellFormed`, typed degree-zero through degree-two coboundaries, `CommittedCochainWellFormed`, `RepresentsExactOneCochain`, `IsOneCocycle`, `oneCocycle _ of _ committedCochainWellFormed`, `FoundationWithSheafCoherence`, and `coherence_implies_structural_preconditions`.

`Axioms/Grassmannian.lean`.
Sections 4 and 6 translate the Grassmannian metric and geometry objects: `GrassmannianDistance`, `chordal_leq_geodesic`, `KarcherMean`, `SheafLaplacian`, `CechNerve`, `PluckerEligibility`, `ConsensusGeometry`, and `ConsensusGeometryWellFormed`.

`Axioms/SpectralBounds.lean` and `Gates/Spectral.lean`.
Section 4 translates the constants $\sqrt{2}/2$ and $2\sqrt{2}$, the predicates `SpectralGapInPoVBounds`, `NearPoVSpectralBoundary`, `TsirelsonPredicate`, `NearTsirelsonBoundary`, the aliases `SpectralGapGate` and `NearSpectralBoundary`, the geometry witness `ByConstructionPoVGeometry`, and the theorems `byConstruction_gap_in_bounds` and `byConstruction_passes_spectral_gate`.

`Tsirelson.lean`.
Section 4 translates the Grassmannian/Laplacian family `Grassmannian_Laplacians`, `Ĩ_G`, `LaplacianSpace`, `SpectralGap`, `Ĩ_1`, `projector_map`, `Classical_Strategy`, `Quantum_Strategy`, and `MaxEigenvalue`; the infinitude and injectivity results `R_infinite`, `infinite_lines`, `Grassmannian_Laplacian_injective`, `projector_injective`, `laplacian_injective`, `Ĩ_1_infinite`, `R_uncountable`, `Ĩ_1_infinite_v2`, and `Ĩ_1_Nonempty`; the Laplacian-space inclusions `laplacian_properties`, `laplacian_is_hermitian`, `Ĩ_1_subset_LaplacianSpace`, `Ĩ_1_subset_LaplacianSpace_v3`, `Ĩ_1_subset_LaplacianSpace_v5`, `Ĩ_1_subset_LaplacianSpace_explicit_v2`, and `Ĩ_1_subset_LaplacianSpace_explicit`; the classical/quantum statement forms `Classical_Bound_Statement` and `Quantum_Bound_Statement`; and the empty-family sanity checks `Classical_Bound_empty_family` and `Quantum_Bound_empty_family`.

`Axioms/Heyting.lean` and `Axioms/OracleModality.lean`.
Section 4 translates the three-valued Heyting algebra, meet, join, implication, negation, failure of double-negation elimination, abstract Heyting collapse, nonfaithful public truth, spectral judgments, consensus actions, oracle queries and answers, Ω_{nv}, witness collapses, quotient equivalence, sheafification, oracle

forcing, η_1/η_2 oracle modalities, Lawvere–Tierney idempotence, boundary judgment behavior, canonical consensus admissibility, oracle pipelines, and the oracle/consensus bridge theorems.

`Axioms / StageAdmissibility . lean`, `Gates / Stage . lean`, and `Gates/Oracle.lean`.
This appendix records the stage and oracle aliases: pipeline stages, stage admissibility, transaction-critical invariant bundles, Python-stage snapshots, refinement predicates, and the theorem that a refined transaction-critical pass is admissible.

B.4 Laplacian Geometry And Search Spaces

`Laplacian/Hodge.lean` and `Laplacian.lean`.
Sections 2–3 translate harmonic projection, residual vector, residual norm, harmonicity, projection membership, nonnegativity, and all equivalences between zero residual, membership in the harmonic subspace, and harmonicity.

`Laplacian/Projection.lean`.
Section 3 translates the projection Laplacian $\Delta_W = I - P_W$ and its pointwise application formula.

`Laplacian/Spectral.lean`.
Section 4 translates sheaf Laplacians, spectral fingerprints, spectral equivalence, scalar Laplacians, kernel and eigenvalue lemmas, scalar fingerprints, homotopy-equivalent Laplacians, and the theorem that spectrum is not homotopy invariant.

`Laplacian/Likelihood.lean`.
Section 4 records the classical and quantum adversary shapes, valid quantum adversaries, Gaussian likelihoods, minimum spectral gaps, normalized success probabilities, optimal-adversary likelihood, and the Gaussian factorization lemmas as spectral-statistical surfaces rather than HCP hardness assumptions.

`Laplacian/Grassmannian.lean`, `Laplacian/Grassmannian/Injectivity.lean`, `Laplacian / Grassmannian / SearchSpace . lean`, and `Laplacian/Grassmannian/ParametricFamily.lean`.
Sections 4 and 6 translate harmonic subspaces, the map to projection Laplacians, the families Ω_r and Ω_1, membership in Laplacian space, nonemptiness of Ω_k, injectivity of the Grassmannian Laplacian map, and the parametric construction of families of k-planes.

B.5 Commitments, Public Artifacts, And TCB Boundary

`Axioms/CochainCommitment/Payload.lean`, `Axioms/CochainCommitment/Blinded.lean`, `Axioms / CochainCommitment / Security . lean`, and `Axioms/CochainCommitment.lean`.
Section 5 translates the canonical modulus, byte lengths, wire length,

`CochainCommitmentData`, edge-count matching, nonempty edge data, supported modulus, well-formedness, commitment sums, common-topology and additive compatibility, zero and addition operations, payload lemmas, binding/hiding/additivity/security axioms, the typed exact-payload to Hodge-blinded-value bridge, exact–harmonic orthogonality and decomposition theorems, blinded and vector commitment axioms, cohomological MAC axioms, and the canonical commitment construction from foundational data.

`TCB/ProofCarrying.lean`, `TCB/Runtime.lean`, `TCB/SNARK.lean`, `TCB/CompressionBoundary.lean`, `TCB/ObservationBoundary.lean`, and `TCB/WitnessVerifier.lean`.
Section 5 and this appendix record hash words, hashed committed cochains, proof-carrying cochains, runtime specification boundaries, runtime acceptance predicates, SNARK-carrying cochains, compression contracts, compressed public data, observation boundaries, gap-decision certificates, witness-compression envelopes, public state updates, and theorems stating that accepted artifacts expose well-formed cochain and oracle predicates.

`Axioms/ZKP.lean`.
Section 5 records the zero-knowledge proof object and its well-formedness predicate as an assumed proof-carrying artifact interface.

B.6 Projection Chain, Proximity Gates, And RS/MCA Endpoint

`Proximity/Problem.lean`, `Proximity/Public.lean`, `Proximity/Projection.lean`, `Proximity/PluckerShadow.lean`, `Proximity/GrassmannianPlucker.lean`, and `Proximity/PluckerCochain/PublicBinding.lean`.
Section 6 records the problem-facing public projection vocabulary, the abstract Plucker coordinate-system and public-shadow contracts, and the interfaces that prepare the concrete Plucker coordinate and finite-code datum.

`Proximity/Grassmannian.lean`, `Proximity/GrassmannianPlucker.lean`, and `Proximity/GrassmannianPlucker/Concrete.lean`.
Section 6 translates harmonic Grassmannian points, exterior coordinates, frame witnesses, decomposability, framed systems, raw and normalized Plucker coordinates, soundness, projective equivalence, projective normalization, projective framed systems, and projective sound inputs.

`Proximity/CochainBridge.lean`, `Proximity/PluckerCochain.lean`, `Proximity/PluckerCochain/Concrete.lean`, and `Proximity/PluckerCochain/PublicBinding.lean`.
Section 6 translates valid frames, Plucker-to-cochain instances, local syndrome targets, codeword predicates, evaluation projections, invariant checklists, payload encoders, commitment-sum residuals, payload realizations, hashed Plucker coordinates, observed frame bindings, and the theorems that valid

Plucker coordinates bind to sheaf-coherent exact one-cochains, hence to accepted committed cocycles and local codewords.

Proximity / CochainPlane . lean, Proximity / CochainPlane / Core . lean, Proximity/CochainPlane/Concrete.lean, Proximity/CochainPlane/ReedSolomon.lean, and Proximity/CochainReedSolomon.lean.
Section 6 translates plane projections, plane extraction contexts, local nondegeneracy, alignment, projected planes, plane targets, plane-to-evaluation projections, typed restriction and syndrome naturality at the cochain/plane and plane/RS transitions, local-codeword-to-target-codeword theorems, separate plane and RS metric handoffs, faithful-on-region plane encodings, interpretations, and RS received words.

Proximity/ProjectionChain.lean and Proximity/ProjectionChain/Core.lean.
Section 6 translates the complete projection chain, including accessors for committed cochain, plane projection, received word, and the composite evaluation projection.

Proximity / Gates . lean, Proximity / Gates / Stage . lean, Proximity/Gates/Final.lean, and Proximity/GateEvidence.lean.
Section 6 translates only proximity gates: source metric, Plucker metric, semantic alignment, optional syndrome adequacy, projective-coordinate respect, three-stage noncollapse through cochain and plane radii, far-region separation, runtime binding, structural bundles, hard-gate and derived naturality certificates, calibrated final certificates, nested witness-local state buses, stage checklists, stage certificates, radius calibrations, final readiness, evidence slots, evidence databases, and the binding/completeness lemmas for proximity evidence.

Proximity / RS / ListDecoding . lean, Proximity / RS / ReedSolomon . lean, Proximity/RS/CorrelatedAgreement.lean, Proximity/RS/Target.lean, and Proximity/RS.lean.
Section 6 translates words, block codes, Hamming and relative Hamming balls, list-size bounds, interleaved and additive codes, RS parameters, vector codewords, correlated-agreement and mutual-correlated-agreement predicates, bad-parameter sets, bridge conditions, calibrated $HomotopicalEncryption$-to-RS bridges, and pullback-correctness theorems.

Proximity/CanonicalRS/Parameters.lean, Proximity/CanonicalRS/PowerDomain.lean, Proximity/CanonicalRS/Encoding.lean, Proximity/CanonicalRS/Metric.lean, Proximity/CanonicalRS/Target.lean, Proximity/CanonicalRS/FarSeparation.lean, Proximity / CanonicalRS / BridgeData . lean, Proximity / CanonicalRS / BridgeWitness . lean, Proximity / CanonicalRS / BridgeCertificate . lean, Proximity / CanonicalRS / CitedBounds . lean, Proximity/CanonicalRS/MCA.lean, Proximity/CanonicalRS/MCA/Line.lean, and Proximity/CanonicalRS/MCA/Concrete.lean.
Section 6 translates the canonical finite field, length, degree bound, rate,

Johnson and capacity radii, smooth domains, power-root domain, canonical encoding, the proof that the unbounded integer encoding is not globally injective, Hamming-to-code metrics, target bounds, far-separation witnesses, bridge data and theorem slots, bridge certificates, cited BCHKS-style bounds, MCA block code, challenge and bad-challenge fractions, and the bound on the projected bad-challenge fraction.

B.7 Subsequent Protocol And Implementation Boundary

`Axioms/Consensus.lean`, `Axioms/Ledger.lean`, `Axioms/Coregulator.lean`, and `Axioms/PythonHomotopyRefinement.lean`.
This appendix records consensus, ledger, coregulator, and Python homotopy refinement as boundary objects. Their full protocol semantics belong to the subsequent protocol paper, but their predicates are named here so the Paper 1 mathematical objects have explicit downstream interfaces.

`Consensus/Geometry.lean`, `Consensus/Reliability.lean`, and `Consensus/ProofOfValue.lean`.
This appendix records Plucker eligibility and consensus geometry aliases, reliability evidence and bounded reliability scores, proximity quality, Proof-of-Value gates, Proof-of-Value inputs, scores with and without reliability/proximity, and the score bounds. These are protocol-facing uses of the Section 4 and Section 6 mathematical predicates.

`Chain.lean`.
This appendix records validators, transactions, blocks, protocol states, spectral bounds, topology and freshness predicates, block validity, tips, heights, extension, transaction and block acceptance, ledger-state updates, key-image preservation theorems, validator eligibility, cumulative gap, preference, transaction composition and identities, quadratic weight, Sybil-resistance weighting, chain-growth monotonicity, fork-resolution transitivity, and connectedness of eligible validators. The detailed protocol architecture remains in the subsequent protocol paper.

References

[AB26] Danel Ahman and Andrej Bauer. Sheaves as oracle computations. *arXiv preprint*, 2026.

[ABF26] Gal Arnon, Dan Boneh, and Giacomo Fenzi. Open problems in list decoding and correlated agreement, 2026. April 8, 2026.

[BSCH+26] Eli Ben-Sasson, Dan Carmon, Ulrich Haböck, Swastik Kopparty, and Shubhangi Saraf. On proximity gaps for reed–solomon codes.

In *Proceedings of the 58th Annual ACM Symposium on Theory of Computing (STOC 2026)*, pages 1157–1167, 2026.

[Hel19] Adam Helfer. First-order homotopical logic. *arXiv preprint*, 2019.

[HG18] Jakob Hansen and Robert Ghrist. Toward a spectral theory of cellular sheaves. *arXiv preprint*, 2018.

[SKdS25] Melanie Swan, Takashi Kido, and Renato P. dos Santos. Higher categorical cryptoeconomics: HoTT blockchains, petri net computad ledgers, coregulator DAOs, and cohomology ZKPs, 2025. Working citation; author affiliations listed in project notes as University College London, Teikyo University, and academicum.ai. Replace with venue metadata before submission.

[Tsi80] Boris S. Tsirelson. Quantum generalizations of bell's inequality. *Letters in Mathematical Physics*, 4:93–100, 1980.